Lab Manual
Electronic Structure of Materials (ESM)

Department of Physics

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Instructions

- Students must complete each experiment before the next session.
- Maintain a proper record of inputs and outputs.
- Include plots, graphs, and analysis wherever required.
- Viva will be conducted based on performed experiments.

%

Chapter 1

Experiment 1: Linux Environment, Vim Editor, and Job Execution

1.1 Aim

To develop working proficiency in the Linux environment, master file editing using Vim, and learn how to execute computational jobs using terminal-based workflows.

1.2 Background Theory

Most electronic structure calculations (DFT, band structure, DOS) are performed on Linux systems, especially in high-performance computing (HPC) environments.

Unlike graphical operating systems, Linux relies heavily on command-line interfaces (CLI), which provide:

- Faster execution
- Better control over processes
- Compatibility with parallel computing

Vim is a powerful, lightweight text editor used for editing input files required by simulation software such as Quantum ESPRESSO.

Understanding Linux and Vim is essential because:

- Input files are manually written
- Jobs are executed via terminal
- Debugging is done through output logs

1.3 Computational Environment

- Operating System: Ubuntu Linux
- Shell: Bash
- Editor: Vim
- Parallel Execution Tool: MPI (mpirun)

1.4 Part A: Basic Linux Commands

Navigation

```
pwd        # show current directory
ls         # list files
cd folder  # change directory
cd ..      # move up
```

File Operations

```
mkdir test_dir     # create directory
rm file.txt        # delete file
cp file1 file2     # copy file
mv file1 file2     # rename/move
```

Viewing Files

```
cat file.txt
less file.txt
head file.txt
tail file.txt
```

1.5 Part B: Vim Editor (Essential Training)

Opening a File

```
vim filename
```

Modes in Vim

- Normal Mode (default)
- Insert Mode (press `i`)
- Command Mode (press `:`)

Basic Commands

```
i        # insert mode
Esc      # back to normal mode
:w       # save
:q       # quit
:wq      # save and quit
:q!      # quit without saving
```

Navigation

```
h j k l   # left, down, up, right
gg        # go to top
G         # go to bottom
```

Editing

```
dd        # delete line
yy        # copy line
p         # paste
u         # undo
```

Search and Replace

```
/text             # search
:%s/old/new/g     # replace all
```

Useful Settings

```
:set nu       # line numbers
:set rnu      # relative numbers
```

1.6 Part C: Connecting to Remote Server (SSH)

Introduction

In computational physics, calculations are often performed on remote servers or high-performance computing (HPC) clusters rather than local machines.

Secure Shell (SSH) is a protocol that allows users to connect to a remote system securely through the terminal.

Basic SSH Command

```
ssh username@server_address
```

Example:

```
ssh student@192.168.1.10
```

Steps to Connect

1. Open terminal on your local machine.
2. Type the SSH command with your username and server IP.
3. Press Enter.
4. Enter password when prompted.

First-Time Connection

When connecting for the first time, you may see:

```
Are you sure you want to continue connecting (yes/no)?
```

Type `yes` and press Enter.

After Login

Once connected:

- You are now working on the remote machine
- All commands will execute on the server

File Transfer (Optional but Useful)

```
scp file.txt username@server_address:/path/to/destination
```

Example:

```
scp scf.in student@192.168.1.10:/home/student/
```

Logout from Server

```
exit
```

1.7 Part D: Job Execution in Terminal

Serial Execution

```
pw.x < scf.in > scf.out
```

Parallel Execution (Multi-core)

```
mpirun -np 4 pw.x < scf.in > scf.out
```

Explanation:

- `-np 4` → number of CPU cores
- Faster computation using parallel processing

Background Jobs

```
pw.x < scf.in > scf.out &
```

Checking Running Jobs

```
top
ps -ef | grep pw.x
```

1.8 Expected Learning Outcomes

After this experiment, students should be able to:

- Navigate Linux file systems confidently
- Edit input files using Vim efficiently
- Execute simulation jobs (serial and parallel)
- Monitor and manage running processes

1.9 Analysis and Discussion

Mastery of Linux and Vim significantly reduces human errors in computational physics workflows.

Efficient use of:

- Vim → reduces syntax errors in input files
- Linux commands → improves workflow speed
- Parallel execution → reduces computation time

This forms the foundation for advanced simulations such as SCF, NSCF, and band structure calculations.

1.10 Common Errors and Pitfalls

- Editing files without saving changes
- Running jobs in wrong directory
- Incorrect file names in commands
- Using too many cores causing system overload
- Syntax errors in input files due to improper editing

1.11 Practice Exercises (Compulsory)

1. Create a directory and navigate inside it.
2. Create a file using Vim and write 10 lines of text.
3. Search and replace a word in the file.
4. Run a dummy job using terminal commands.
5. Execute a command using 2 cores and compare time.

Chapter 2

Experiment 2: Crystal Visualization using VESTA

2.1 Aim

To create crystall structure using following step-by-step procedure, visualize them and export for furthure DFT calculation using VESTA software .

2.2 Theory

VESTA (Visualization for Electronic and STructural Analysis) is a 3D visualization program used for structural modeling, visualization, and analysis of crystal structures. It allows users to create, manipulate, and export crystal structures in various file formats including VASP POSCAR (.vasp), CIF, and XYZ.

2.3 Step-by-Step Flowchart for Conventional Fe

The conventional cell of Iron exhibits Body-Centered Cubic (BCC) structure with space group *Im-3m* (No. 229). The unit cell parameters are $a = b = c = 2.87$ Å and $\alpha = \beta = \gamma = 90°$. The BCC structure contains two atoms per conventional cell: one at the corner (0,0,0) and one at the body center (0.5,0.5,0.5).

```
START: Open VESTA
↓
File → New Structure
        | Rename to:  Fe_Conventional
↓
Click "Unit Cell" Tab
        | Set a = b = c = 2.87 Å
        | Set α = β = γ = 90°
↓
Click "Select Space Group"
        | Search or type:  229
        | Select:  Im-3m (No.  229)
        | Click OK
↓
Click "Structure Parameters" Tab
↓
Click "Add" Button (for first atom)
        | Element:  Fe
        | x = 0, y = 0, z = 0
        | Occupancy = 1
        | Click OK
↓
Click "Add" Again (for second atom)
        | Element:  Fe
        | x = 0.5, y = 0.5, z = 0.5
        | Occupancy = 1
        | Click OK
↓
Click "OK" on Structure Parameters Window
↓
View Structure:
        | Mouse Left Button:  Rotate
        | Mouse Right Button:  Zoom
        | Mouse Middle Button:  Pan
↓
Optional:  Add Bonds
        | Edit → Bonds → Search Bonds → OK
↓
Save/Export:
        | File → Save (as .vesta for later editing)
        | File → Export Data → VASP → Save as POSCAR_Fe_conv
↓
END: Structure ready
```

Figure 2.1: Command chain to create conventional cell of Fe (BCC) in VESTA

2.4 General Procedure to Create Any Structure in VESTA

The formation of any crystal structure involves following three primary steps:

2.4.1 Step-by-Step Visual Guide



1. **Click** File → New Structure



2. **Rename** the structure appropriately: `Fe_Conventional`, `Cu_FCC`, `NaCl`

1. Click **Unit Cell** tab
2. Enter lattice parameters:
   a, b, c (in Å) and α, β, γ (in degrees)

   Example for Fe: `a = 2.87 Å, b = 2.87 Å, c = 2.87 Å`
   $\alpha = 90°$, $\beta = 90°$, $\gamma = 90°$
3. Click **Select Space Group** button
4. Choose required space group (e.g., `229` → `Im-3m`)
5. Click **OK**

1. Click **Structure Parameters** tab
2. Click **Add** button for each atom
3. Provide the following details:

   Element: Fe, Cu, C, Na, Cl
   Fractional coordinates: x, y, z (0 to 1)
   Occupancy: 1
4. Click **OK** after entering all atoms

Atom No.	Element	x	y	z
1	Fe1	0	0	0
2	Fe	0.5	0.5	0.5

These coordinates represent corner and body-centered positions in BCC lattice

2.5 Structures Created in this Experiment

A total of six structures were created following the above procedure:

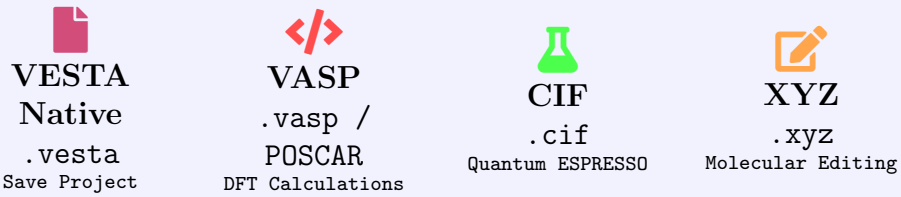
S.No.	Structure	Space Group	Lattice Parameters (Å)	Atomic Coordinates (Fractional)
1	Conventional Fe (BCC)	*Im-3m* (229)	$a = 2.87$	Fe: (0,0,0), (0.5,0.5,0.5)
2	Primitive Fe	*R-3m* (166)	$a = 2.87$	Fe: (0,0,0)
3	Conventional Cu (FCC)	*Fm-3m* (225)	$a = 3.61$	Cu: (0,0,0), (0,0.5,0.5), (0.5,0,0.5), (0.5,0.5,0)
4	Primitive Cu	*R-3m* (166)	$a = 3.61$	Cu: (0,0,0)
5	Graphene	*P6/mmm* (191)	$a = b = 2.46$, $c = 15$, $\gamma = 120°$	C: (0,0,0.5), (1/3,2/3,0.5)
6	NaCl	*Fm-3m* (225)	$a = 5.64$	Na: (0,0,0), (0,0.5,0.5), (0.5,0,0.5), (0.5,0.5,0)
				Cl: (0.5,0,0), (0,0.5,0), (0,0,0.5), (0.5,0.5,0.5)

Table 2.1: Summary of all six crystal structures created in VESTA

2.6 Saving and Exporting Structures

2.6.1 Saving and Exporting Structures from VESTA

Export Options in VESTA



1. **File → Save**
2. Saves as `.vesta` file
3. **Preserves:** All visualization settings (bonds, colors, view angles, lighting)
4. **Use when:** You want to continue editing the structure later

1. **File → Export Data**
2. Choose **VASP** as file type
3. Save with filename like `POSCAR_Fe_conv`
4. **Use for:** DFT calculations in VASP software

POSCAR File Format

```
Fe_Conventional
1.0
2.87 0.00 0.00
0.00 2.87 0.00
0.00 0.00 2.87
Fe
2
Direct
0.0 0.0 0.0
0.5 0.5 0.5
```

1. **File → Export Data**
2. Choose **CIF** (Crystallographic Information File) as file type
3. Save with filename like `Fe_conv.cif`
4. **Use for:** Quantum ESPRESSO, Materials Studio, and crystal structure databases

1. **File → Export Data**
2. Choose **XYZ** as file type
3. Save with filename like `Fe_conv.xyz`
4. **Use for:** Further editing in Avogadro, VMD, OVITO, or manual text editing

XYZ File Format

```
2
Fe Conventional Cell
Fe 0.000 0.000 0.000
Fe 1.435 1.435 1.435
```

Quick Tip
Always save a `.vesta` file first, then export to other formats.
This preserves your work for future editing!

2.7 Structures Created in this Experiment

A total of **15 structures** were created following the above procedure (6 required + 9 additional):

S.No.	Structure	Space Group	Lattice Parameters (Å)	Atomic Coordinates (Fractional)
1	Conventional Fe (BCC)	*Im-3m* (229)	$a = 2.87$	Fe: (0,0,0), (0.5,0.5,0.5)
2	Primitive Fe	*R-3m* (166)	$a = 2.87$	Fe: (0,0,0)
3	Conventional Cu (FCC)	*Fm-3m* (225)	$a = 3.61$	Cu: (0,0,0), (0,0.5,0.5), (0.5,0,0.5), (0.5,0.5,0)
4	Primitive Cu	*R-3m* (166)	$a = 3.61$	Cu: (0,0,0)
5	Graphene	*P6/mmm* (191)	$a = b = 2.46$, $c = 15$, $\gamma = 120°$	C: (0,0,0.5), (1/3,2/3,0.5)
6	NaCl	*Fm-3m* (225)	$a = 5.64$	Na: (0,0,0), (0,0.5,0.5), (0.5,0,0.5), (0.5,0.5,0) Cl: (0.5,0,0), (0,0.5,0), (0,0,0.5), (0.5,0.5,0.5)

Table 2.2: Summary of all six crystal structures created in VESTA

2.7.1 Additional Structures

The following 9 structures were also built using VESTA software:

S.No.	Structure	Type
1	SiC	Conventional cell
2	SiC	Primitive cell
3	Si	Diamond type cell
4	C	Diamond type cell
5	Si	Conventional cell
6	C	Conventional cell
7	ZnS	Conventional cell
8	ZnS	Hexagonal cell
9	$PtSe_2$	Bi-layer

Table 2.3: Additional structures created in VESTA

2.8 Result

All six crystal structures (Conventional Fe, Primitive Fe, Conventional Cu, Primitive Cu, Graphene, and NaCl) were successfully created and visualized using VESTA. The structures were saved as `.vesta` files and exported as VASP POSCAR files for future use.

Important Note: Adding Vacuum for 2D Structures

For 2D structures (Graphene, $PtSe_2$, MoS_2), always set $c = 15$ **to** 20 **Å** in the Unit Cell tab to add vacuum in the z-direction. This prevents artificial interaction between periodic layers in DFT calculations.

2.9 Precautions

1. Always use fractional coordinates, not Cartesian coordinates
2. Ensure correct space group is selected before adding atoms
3. For 2D materials like graphene, add sufficient vacuum along c-axis
4. Save files frequently to avoid data loss

2.10 Viva Questions

1. What is the difference between conventional and primitive cell?
2. How many atoms are present in BCC conventional cell?
3. What is the space group of NaCl structure?
4. Why do we add vacuum for graphene structure?
5. What is the purpose of exporting as VASP POSCAR file?

Chapter 3

Lab 3: Quantum ESPRESSO DFT Code and SCF Calculation (Self-Consistent Field Calculation)

3.1 Aim

To perform a Self-Consistent Field (SCF) calculation using Quantum ESPRESSO, using the CIF file generated from VESTA in Chapter 2.

3.2 Theory

3.2.1 What is DFT?

Density Functional Theory (DFT) is a quantum mechanical method used to investigate the electronic structure of many-body systems (atoms, molecules, solids). It calculates the ground state properties of a system using electron density rather than wavefunctions.

3.2.2 Popular DFT Codes

Several DFT software packages are available for materials science calculations:

Code	Developer	License	Best For
Quantum ESPRESSO	Open source	GPL	Materials science, solids
VASP	University of Vienna	Commercial	High-throughput DFT
CASTEP	University of Cambridge	Commercial	Solids, surfaces
ABINIT	Open source	GPL	Molecular dynamics
GPAW	Open source	GPL	Real-space DFT

Table 3.1: Popular DFT software packages

3.2.3 Quantum ESPRESSO

Quantum ESPRESSO (opEn Source Package for Research in Electronic Structure, Simulations, and Optimization) is an integrated suite of open-source computer codes for electronic-structure calculations and materials modeling.

Quantum ESPRESSO Resources

- **Official Website**: `https://www.quantum-espresso.org`
- **Documentation/Manual**: `https://www.quantum-espresso.org/resources/documentati`
- **Pseudopotential Library**: `https://www.quantum-espresso.org/pseudopotentials`
- **Input Generator**: `https://www.materialscloud.org/work/tools/qeinputgenerator`

3.2.4 Hierarchy of DFT Calculations

Before performing DFT calculations, it is important to understand the sequence of calculations:

Workflow of DFT Calculations

Step 1: Relaxation (Structure Optimization)
↓
Step 2: SCF (Self-Consistent Field)
↓
Step 3: NSCF (Non-Self-Consistent Field)
↓
Step 4: Band Structure & DOS
↓
Step 5: Further Properties (Optical, Phonon, etc.)

1. Relaxation (Structure Optimization)

What it does: Finds the equilibrium atomic positions and lattice parameters by minimizing forces and stress.

Why needed: Experimental structures may not be at their ground state. Relaxation ensures the structure is stable.

Input parameters: `calculation = 'relax'` or `'vc-relax'` (variable cell)

Output: Optimized coordinates and lattice parameters.

2. SCF (Self-Consistent Field Calculation)

What it does: Solves Kohn-Sham equations iteratively until input and output potentials converge.

Why needed: Provides ground state electron density and total energy.

Input parameters: `calculation = 'scf'`

Output: Total energy, Fermi energy, converged charge density.

3. NSCF (Non-Self-Consistent Field Calculation)

What it does: Performs a single diagonalization using fixed potential from SCF.

Why needed: Calculates eigenvalues on a denser k-point grid for band structure and DOS.

Input parameters: `calculation = 'nscf'`

Output: Kohn-Sham eigenvalues on a fine k-point mesh.

4. Band Structure & DOS

What it does:

- **Band Structure:** Plots E vs. k along high-symmetry paths in Brillouin zone.
- **DOS (Density of States):** Shows number of states per energy interval.

Why needed: Determines if material is metal, semiconductor, or insulator. Band gap can be calculated.

Tools used: `bands.x`, `dos.x` (Quantum ESPRESSO), or `pyprocar`, `sumo`

5. Further Calculations

Examples:

- **Phonon calculations** (`ph.x`): Lattice dynamics, thermal properties
- **Optical properties** ($\epsilon.x$): Dielectric function, absorption
- **Molecular dynamics** (`cp.x`): Finite temperature behavior
- **DFT+U**: For strongly correlated materials
- **Hybrid functionals** (HSE): For accurate band gaps

3.2.5 Connection to Previous Chapter

The `.cif` (Crystallographic Information File) exported from VESTA in Chapter 2 will be directly uploaded to the Quantum ESPRESSO Input Generator. This creates a seamless workflow from structure visualization to DFT calculation.

Complete Workflow: Chapter 2 → Chapter 4 → Beyond

VESTA (Chapter 2)
↓
`Export Data → CIF`
↓
Quantum ESPRESSO Input Generator
↓
Relaxation (if needed) → `calculation = 'relax'`
↓
SCF Calculation → `calculation = 'scf'`
↓
NSCF Calculation → `calculation = 'nscf'`
↓
Band Structure & DOS → `bands.x, dos.x`
↓
Further Properties (Optical, Phonon, etc.)

3.3 Step-by-Step Procedure for SCF Calculation

Step 1: Open Terminal & Create Directory

1. Open the **Terminal** (Ctrl+Alt+T in Linux)
2. Use the `mkdir` command to create a new directory:

```
mkdir scf_calculation
```

3. Navigate into the project folder:

```
cd scf_calculation
```

Step 2: Create Project Folder Structure

1. Create folders for pseudopotentials and outputs:

```
mkdir pseudo outputs
```

2. Copy the CIF file from Chapter 2 to this folder:

```
cp /path/to/chapter2/Fe_conv.cif ./
```

Step 3: Upload CIF to Quantum ESPRESSO Input Generator

1. Open **Quantum ESPRESSO Input Generator** website:

   `https://www.materialscloud.org/work/tools/qeinputgenerator`

2. **Upload the CIF file** exported from VESTA:
   - Click **"Upload Structure"**
   - Select your `.cif` file (e.g., `Fe_conv.cif`, `NaCl.cif`, `Graphene.cif`)
   - The generator automatically reads:
     - Lattice parameters (a, b, c, α, β, γ)
     - Space group
     - Atomic coordinates
     - Atomic species
3. Set **Calculation Type** to `SCF`
4. Set other parameters:
   - `ecutwfc` = 50 Ry (plane-wave cutoff)
   - `k-points` = 8x8x8 (for bulk) or 8x8x1 (for 2D materials)
5. Download the generated input file as `material_scf.in`

Note

The CIF file contains all structural information. You do NOT need to manually enter lattice parameters or atomic coordinates!

Step 4: Get Pseudopotential from Quantum ESPRESSO

1. Download pseudopotentials from SSSP (Standard Solid State Pseudopotentials):

   `https://www.materialscloud.org/discover/sssp/`

2. Search for your element (e.g., Fe, Cu, C, Na, Cl)

3. Download the `.upf` file (e.g., `Fe.pbe-spn-kjpaw.UPF`)

4. Save pseudopotentials in the `pseudo/` folder:

```
mv  /Downloads/Fe.pbe*.UPF ./pseudo/
```

Step 5: Save Input File and Modify Pseudopotential Paths

1. Save the downloaded input file as `material_scf.in` in your project folder
2. Open the file using any text editor:

```
nano material_scf.in
```

3. Modify the following:
   - **pseudo_dir** = './pseudo/'
   - Change `material.pbe.upf` to your actual pseudopotential filename

Example: Modified Input File (`Fe_scf.in`)

```
&CONTROL
calculation = 'scf',
prefix = 'Fe',
outdir = './outputs/',
pseudo_dir = './pseudo/',
/
&SYSTEM
ibrav = 0, !  Reads from CIF
nat = 2,
ntyp = 1,
ecutwfc = 50,
/
&ELECTRONS
conv_thr = 1e-8,
/
ATOMIC_SPECIES
Fe 55.845 Fe.pbe-spn-kjpaw.UPF
CELL PARAMETERS {angstrom}
2.87 0.00 0.00
0.00 2.87 0.00
0.00 0.00 2.87
ATOMIC_POSITIONS (crystal)
Fe 0.0 0.0 0.0
Fe 0.5 0.5 0.5
K_POINTS (automatic)
8 8 8 0 0 0
```

3.4 Complete Workflow Diagram

```
START: From Chapter 2 (VESTA)
↓
CIF file exported from VESTA
        | Example:  Fe_conv.cif, NaCl.cif, Graphene.cif
↓
Step 1:  Open Terminal
        | mkdir scf_calculation
        | cd scf_calculation
↓
Step 2:  Create folders
        | mkdir pseudo outputs
        | Copy CIF file to this folder
↓
Step 3:  Quantum ESPRESSO Input Generator
        | Upload CIF file
        | Set calculation = scf
        | Download material_scf.in
↓
Step 4:  Get Pseudopotential
        | Download .upf from SSSP
        | Save in pseudo/ folder
↓
Step 5:  Modify Input File
        | Change pseudo_dir = './pseudo/'
        | Update pseudopotential filename
↓
Run SCF Calculation
        | pw.x < material_scf.in > material_scf.out
↓
Check Convergence
        | grep "!" material_scf.out
↓
END: SCF completed successfully
```

Figure 3.1: Complete workflow from VESTA (Chapter 2) to SCF calculation

3.5 Running the SCF Calculation

Once the input file is ready, run the calculation using:

```
pw.x < material_scf.in > material_scf.out
```

3.6 Output Files

After successful completion, the following files are generated:

- `material_scf.out` - Main output file with:
  - Total energy
  - Convergence history
  - Fermi energy
- `outputs/` - Directory containing wavefunctions and charge density

3.7 Checking Convergence

How to Check if SCF Converged

1. Open the output file:

```
grep "convergence" material_scf.out
```

2. Look for:

```
!  total energy = -XXXX.XX Ry
```

3. If you see `convergence has been achieved`, your SCF calculation is successful!

3.8 Summary Table

Step	Action	Output
1	Create directory using mkdir	Project folder
2	Copy CIF from Chapter 2	.cif file
3	Upload CIF to QE Input Generator	.in file
4	Download pseudopotential	.upf file
5	Modify pseudo_dir & filename	Ready .in file
Run	pw.x ¡ .in ¿ .out	.out file with results

Table 3.2: Summary of SCF calculation steps

3.9 Precautions

- Ensure the CIF file from Chapter 2 is correctly exported (File → Export Data → CIF)
- Verify pseudopotential path (`pseudo_dir`) is correct
- Check pseudopotential filename matches exactly with the file in `pseudo/` folder
- For 2D structures (Graphene, $PtSe_2$), use **k-points = 8x8x1** (not 8x8x8)
- Always use vacuum ($c \geq 15$ Å) for 2D materials (as learned in Chapter 2)

3.10 Viva Questions

1. What are the different DFT codes available? Name at least three.

2. What is the official website of Quantum ESPRESSO?
3. Why do we upload the CIF file from VESTA to the input generator?
4. What is the difference between Relaxation, SCF, and NSCF calculations?
5. What is the purpose of an SCF calculation?
6. What are pseudopotentials and why are they needed?
7. What does "convergence" mean in SCF calculations?
8. How do you check if your SCF calculation converged successfully?
9. What is the difference between Band Structure and DOS?
10. What are the next steps after SCF and NSCF?

Chapter 4

Experiment 4: ECUT and k-points Convergence

4.1 Aim

To determine the optimal plane-wave kinetic energy cutoff (ECUT) and k-point grid for accurate total energy calculations with optimized computational time.

where **G** are reciprocal lattice vectors and **k** is the wavevector.

Why ECUT Convergence is Important

- Higher ECUT = More accurate results
- Higher ECUT = More computational time and memory
- Need to find the optimal value where energy becomes independent of ECUT
- Typical range: 30-100 Ry (400-1360 eV) for solids

4.1.1 k-point Convergence

The Brillouin zone integration is performed over a discrete grid of k-points. The k-point grid must be dense enough to accurately sample the electronic structure.

Why k-point Convergence is Important

- More k-points = More accurate total energy
- More k-points = More computational time
- Optimal grid: energy converges with respect to k-points
- For metals, smearing is needed to handle the Fermi surface

4.2 Procedure for ECUT Convergence

Steps for ECUT Convergence

1. Create a bash script with a list of ECUT values
2. For each ECUT value:
   (a) Generate SCF input file with that ECUT
   (b) Run `pw.x` calculation
   (c) Extract total energy from output
   (d) Store ECUT vs Energy data
3. Plot Energy vs ECUT
4. Find the ECUT where energy converges

4.2.1 ECUT Convergence Script

ECUT Convergence Bash Script

```
#!/bin/bash
# --------- Timing Start ---------
SECONDS=0
echo "Started at: $(date)"

# ----------------------------------
# QE run Configurations
# ----------------------------------
QE_CMD="mpirun -np 2 /home/raunak/DFT_code/q-e-qe-7.4.1/bin/pw.x"

# ----------------------------------
# Convergence configurations
#----------------------------------
ENCUT_LIST=(35 45 55 65 75 85 95)

# ----------------------------------
# Store energy file and Generate gnuplot script
# ----------------------------------
ENCUT_ENERGY_FILE="energy.dat"
ENCUT_GNUPLOT_SCRIPT="plot_energy.gnu"

# Clean old files
rm -f $ENCUT_ENERGY_FILE
echo "Starting ENCUT-convergence"

# ----------------------------------
# Loop over ENCUT values
# ----------------------------------
for ENCUT in "${ENCUT_LIST[@]}"; do
    echo "Running for ENCUT = $ENCUT"

    # Create scf.in for current ENCUT
    cat > scf.in <<EOF
&CONTROL
    calculation = 'scf'
    outdir = './out/'
    prefix = 'aiida'
    pseudo_dir = './'
    tprnfor = .true.
    tstress = .true.
    verbosity = 'high'
/
&SYSTEM
    degauss = 2.0000000000d-02
    ecutrho = 720
    ecutwfc = ${ENCUT}
    ibrav = 0
    nat = 4
    nosym = .false.
    ntyp = 1
    occupations = 'smearing'
    smearing = 'cold'
/
&ELECTRONS
    conv_thr = 8.0000000000d-10
    electron_maxstep = 80
    mixing_beta = 4.0000000000d-01
/
ATOMIC_SPECIES
    Cu 63.546 Cu.pbe-dn-kjpaw_psl.1.0.0.UPF
ATOMIC_POSITIONS crystal
    Cu 0.0000000000 0.0000000000 0.0000000000
    Cu 0.5000000000 0.5000000000 0.0000000000
    Cu 0.5000000000 0.0000000000 0.5000000000
    Cu 0.0000000000 0.5000000000 0.5000000000
K_POINTS automatic
    12 12 12 0 0 0
CELL_PARAMETERS angstrom
    3.5774306700 0.0000000000 0.0000000000
    0.0000000000 3.5774306700 0.0000000000
    0.0000000000 0.0000000000 3.5774306700
EOF

    # Run QE
    $QE_CMD < scf.in > scf.out
```

4.2.2 ECUT Convergence Results

ECUT Convergence Result

The optimal kinetic energy cutoff was determined to be:

85 eV (ecutwfc)

Beyond 85 eV, the total energy becomes independent of the cutoff energy.

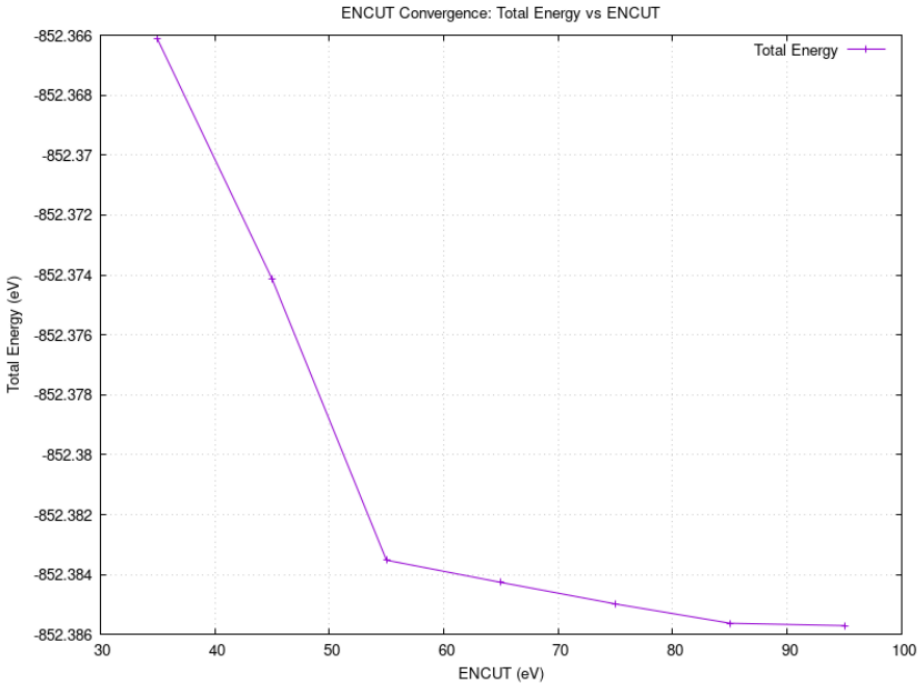


Figure 4.1: ECUT Convergence: Total Energy vs ECUT for Copper

4.3 Procedure for k-point Convergence

Steps for k-point Convergence

1. Set ECUT to optimal value (85 eV)
2. Create a bash script with a list of k-point values (2 to 20)
3. For each k-point value:
   (a) Generate SCF input with `kpoints = N N N`
   (b) Run `pw.x` calculation
   (c) Extract total energy
   (d) Store k-point vs Energy data
4. Plot Energy vs k-points
5. Find the k-point where energy converges

4.3.1 k-point Convergence Script

k-point Convergence Bash Script

```
#!/bin/bash
# --------- Timing Start ---------
SECONDS=0
echo "Started at: $(date)"

# ----------------------------------
# QE run Configurations
# ----------------------------------
QE_CMD="mpirun -np 2 /home/raunak/DFT_code/q-e-qe-7.4.1/bin/pw.x"

# ----------------------------------
# Convergence configurations
#----------------------------------
KPOINT_LIST=(2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20)

# ----------------------------------
# Store energy file and Generate gnuplot script
# ----------------------------------
KPOINT_ENERGY_FILE="energy.dat"
KPOINT_GNUPLOT_SCRIPT="plot_energy.gnu"

# Clean old files
rm -f $KPOINT_ENERGY_FILE
echo "Starting KPOINT-convergence"

# ----------------------------------
# Loop over KPOINT values
# ----------------------------------
for KPOINT in "${KPOINT_LIST[@]}"; do
    echo "Running for KPOINT = $KPOINT"

    # Create scf.in for current KPOINT
    cat > scf.in <<EOF
&CONTROL
    calculation = 'scf'
    outdir = './out/'
    prefix = 'aiida'
    pseudo_dir = './'
    tprnfor = .true.
    tstress = .true.
    verbosity = 'high'
/
&SYSTEM
    degauss = 2.0000000000d-02
    ecutrho = 720
    ecutwfc = 85
    ibrav = 0
    nat = 4
    nosym = .false.
    ntyp = 1
    occupations = 'smearing'
    smearing = 'cold'
/
&ELECTRONS
    conv_thr = 8.0000000000d-10
    electron_maxstep = 80
    mixing_beta = 4.0000000000d-01
/
ATOMIC_SPECIES
    Cu 63.546 Cu.pbe-dn-kjpaw_psl.1.0.0.UPF
ATOMIC_POSITIONS crystal
    Cu 0.0000000000 0.0000000000 0.0000000000
    Cu 0.5000000000 0.5000000000 0.0000000000
    Cu 0.5000000000 0.0000000000 0.5000000000
    Cu 0.0000000000 0.5000000000 0.5000000000
CELL_PARAMETERS angstrom
    3.5774306700 0.0000000000 0.0000000000
    0.0000000000 3.5774306700 0.0000000000
    0.0000000000 0.0000000000 3.5774306700
K_POINTS automatic
    ${KPOINT} ${KPOINT} ${KPOINT} 0 0 0
EOF

    # Run QE
    $QE_CMD < scf.in > scf.out
```

4.3.2 k-point Convergence Results

k-point Convergence Result

The optimal number of k-points was determined to be:

12 (12 x 12 x 12)

Beyond 12x12x12, the total energy becomes independent of k-points.

k-point Convergence Plot for Cu

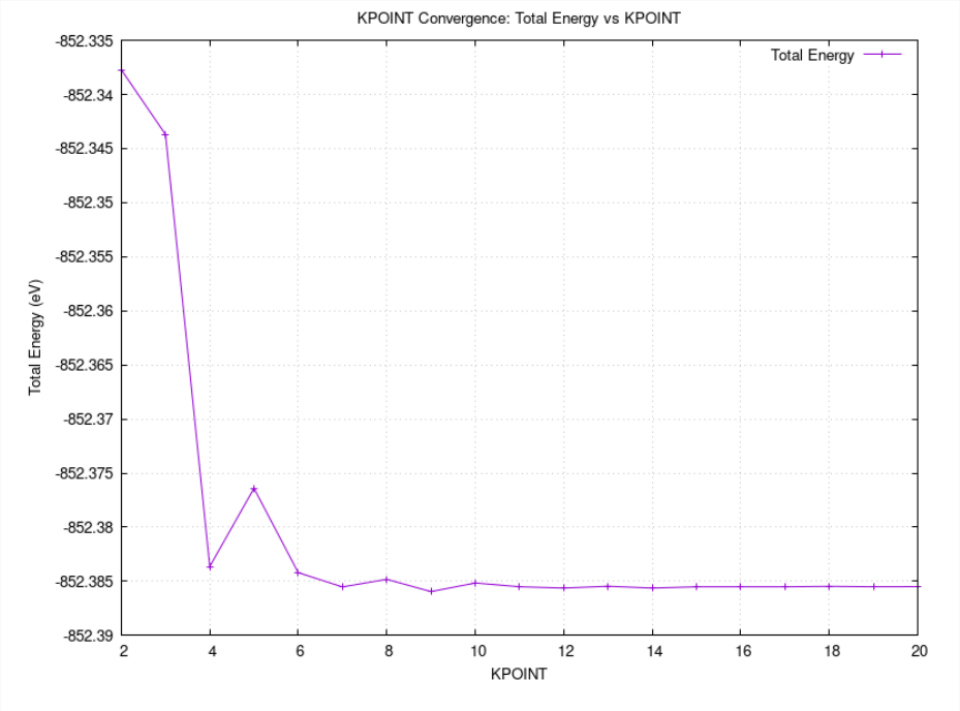


Figure 4.2: k-point convergence plot showing optimal grid at 12x12x12

4.4 Summary of Results

Parameter	Value Tested	Optimal Value
ECUT (eV)	35, 45, 55, 65, 75, 85, 95	**85 eV**
k-points (NxNxN)	2 to 20	**12 x 12 x 12**

Table 4.1: Optimal parameters for Cu SCF calculations

4.5 Importance of Convergence Tests

Why Both Tests are Necessary

- **ECUT**: Determines accuracy of wavefunction expansion
- **k-points**: Determines accuracy of Brillouin zone integration
- Both affect:
  - Total energy accuracy
  - Computational cost
  - Reliability of further properties (DOS, bands, etc.)
- Always perform both tests for a new material
- Optimal values vary for different materials and structures

4.6 Precautions

- Use the same pseudopotential for all ECUT tests
- For k-point convergence, use the optimal ECUT value
- Check for convergence in total energy, not just forces

Chapter 5

Lab 6: NSCF and DOS Calculations

5.1 Aim

To perform Non-Self-Consistent Field (NSCF) calculations and Density of States (DOS) calculations for Copper metal using Quantum ESPRESSO.

5.2 Theory

5.2.1 Non-Self-Consistent Field (NSCF) Calculation

NSCF calculations use the ground state charge density obtained from a previous SCF calculation but compute eigenvalues on a denser k-point grid.

Why NSCF is Needed

- SCF calculations use moderate k-point grids (computational efficiency)
- Band structure and DOS need very dense k-point grids
- NSCF reuses SCF charge density (saves computational time)
- Provides accurate eigenvalues at many k-points

5.2.2 Density of States (DOS)

DOS represents the number of electronic states per unit energy interval. It is a fundamental property for understanding material behavior.

DOS: Key Concepts

- **DOS(E)** = Number of states between E and E+dE
- **Metals**: Finite DOS at Fermi energy
- **Semiconductors**: Gap in DOS near Fermi energy
- **Insulators**: Large band gap
- **Integration methods**:
  - Gaussian smearing (metals)
  - Tetrahedron method (insulators, accurate DOS)

5.3 Connection to Previous Experiments

Workflow from Chapter 4 to Chapter 6

Chapter 4: ECUT & k-point Convergence

↓

Optimal Parameters: ECUT = 85 eV, k-points = 12x12x12

↓

Chapter 5: SCF Calculation

↓

Chapter 6: NSCF & DOS Calculations

5.4 NSCF Input File

NSCF Input File (nscf.in)

```
&CONTROL
    calculation = 'nscf'            ! Non-SCF calculation
    etot_conv_thr = 4.0000000000d-05
    forc_conv_thr = 1.0000000000d-04
    outdir = './out/'
    prefix = 'aiida'
    pseudo_dir = './'
    tprnfor = .true.
    tstress = .true.
    verbosity = 'high'
/
&SYSTEM
    degauss = 2.0000000000d-02
    ecutrho = 7.2000000000d+02
    ecutwfc = 9.0000000000d+01
    ibrav = 0
    nat = 4
    nosym = .false.
    ntyp = 1
    occupations = 'smearing'
    smearing = 'cold'
    nbnd = 26                       ! Number of bands (more than occupied)
/
&ELECTRONS
    conv_thr = 8.0000000000d-10
    electron_maxstep = 80
    mixing_beta = 4.0000000000d-01
/
ATOMIC_SPECIES
    Cu 63.546 Cu.pbe-dn-kjpaw_psl.1.0.0.UPF
ATOMIC_POSITIONS crystal
    Cu 0.0000000000 0.0000000000 0.0000000000
    Cu 0.5000000000 0.5000000000 0.0000000000
    Cu 0.5000000000 0.0000000000 0.5000000000
    Cu 0.0000000000 0.5000000000 0.5000000000
K_POINTS automatic
    12 12 12 0 0 0                  ! Dense k-point grid
CELL_PARAMETERS angstrom
    3.5774306700 0.0000000000 0.0000000000
    0.0000000000 3.5774306700 0.0000000000
    0.0000000000 0.0000000000 3.5774306700
```

Key Differences: SCF vs NSCF

- `calculation = 'nscf'` (instead of 'scf')
- `nbnd = 26` (includes empty bands for DOS)
- Uses same charge density from previous SCF run
- Must point to same `outdir` as SCF

5.5 DOS Workflow

Steps for DOS Calculation

1. Run **SCF** calculation (Chapter 5)
2. Run **NSCF** calculation with dense k-points
3. Run **dos.x** post-processing tool
4. Plot DOS data

5.5.1 DOS Input File (`dos.in`)

DOS Calculation Input (`dos.in`)

```
&CONTROL
    calculation = 'nscf'            ! Same as NSCF
    etot_conv_thr = 4.0000000000d-05
    forc_conv_thr = 1.0000000000d-04
    outdir = './out/'
    prefix = 'aiida'
    pseudo_dir = './'
    tprnfor = .true.
    tstress = .true.
    verbosity = 'high'
/
&SYSTEM
    degauss = 2.0000000000d-02
    ecutrho = 7.2000000000d+02
    ecutwfc = 9.0000000000d+01
    ibrav = 0
    nat = 4
    nosym = .TRUE.
    ntyp = 1
    occupations = 'tetrahedra'      ! Tetrahedron method for DOS
    smearing = 'cold'
/
&ELECTRONS
    conv_thr = 8.0000000000d-10
    electron_maxstep = 80
    mixing_beta = 4.0000000000d-01
/
ATOMIC_SPECIES
    Cu 63.546 Cu.pbe-dn-kjpaw_psl.1.0.0.UPF
ATOMIC_POSITIONS crystal
    Cu 0.0000000000 0.0000000000 0.0000000000
    Cu 0.5000000000 0.5000000000 0.0000000000
    Cu 0.5000000000 0.0000000000 0.5000000000
    Cu 0.0000000000 0.5000000000 0.5000000000
K_POINTS automatic
    12 12 12 0 0 0
CELL_PARAMETERS angstrom
    3.5774306700 0.0000000000 0.0000000000
    0.0000000000 3.5774306700 0.0000000000
    0.0000000000 0.0000000000 3.5774306700
```

Important Changes in DOS Input

- `nosym = .TRUE.` (use full symmetry for accurate DOS)
- `occupations = 'tetrahedra'` (better for DOS than smearing)
- Same `outdir` and `prefix` as SCF

5.5.2 DOS Post-Processing Input (`pp.dos.in`)

DOS Post-Processing Input (`pp.dos.in`)

```
&DOS
    prefix = 'aiida'              ! Must match SCF/NSCF prefix
    outdir = './out'              ! Must match SCF/NSCF outdir
    fildos = 'Cu_dos.dat'         ! Output file name
    emin = -9.0                   ! Minimum energy (eV)
    emax = 16.0                   ! Maximum energy (eV)
/
```

Running DOS Post-Processing

```
mpirun -np 2 /home/raunak/DFT_code/q-e-qe-7.4.1/bin/dos.x <
pp.dos.in > pp.dos.out
```

5.6 Complete Workflow Diagram

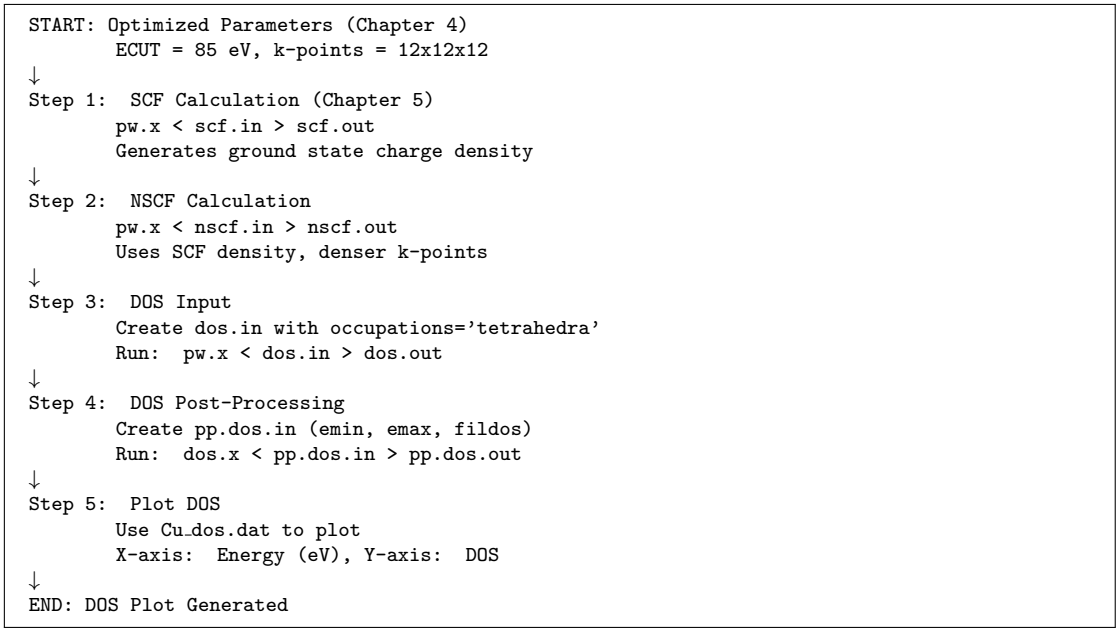


Figure 5.1: Complete workflow for NSCF and DOS calculations

5.7 DOS Output

DOS Plot Description

The DOS plot shows:

- **X-axis**: Energy states (eV)
- **Y-axis**: Density of States (states/eV)
- Fermi energy at 0 eV (usually shifted)
- For Cu (metal): Finite DOS at Fermi level
- Peaks correspond to d-states and sp-states

Schematic DOS Plot for Cu

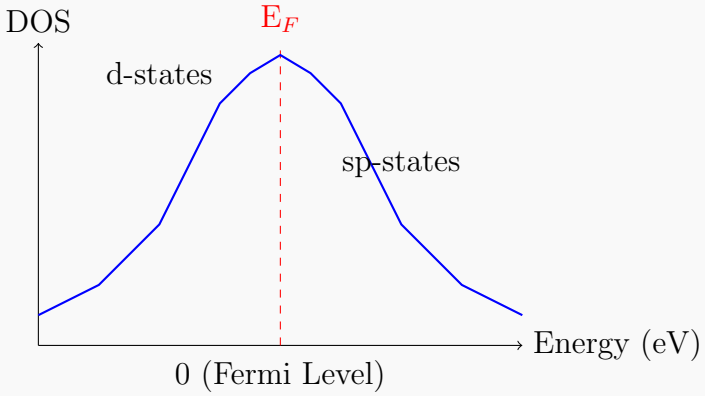


Figure 5.2: Schematic DOS plot for Copper metal

5.8 Summary Table

File	Command	Purpose
`scf.in`	`pw.x < scf.in > scf.out`	Ground state density
`nscf.in`	`pw.x < nscf.in > nscf.out`	Dense k-point eigenvalues
`dos.in`	`pw.x < dos.in > dos.out`	DOS with tetrahedra
`pp.dos.in`	`dos.x < pp.dos.in > pp.dos.out`	Extract DOS data
`Cu_dos.dat`	-	Plot DOS

Table 5.1: Summary of files and commands for DOS calculation

5.9 Key Parameters for DOS

Parameter	NSCF	DOS
calculation	nscf	nscf
occupations	smearing	tetrahedra
nosym	.false.	.TRUE.
nbnd	26	Default
smearing	cold	cold
degauss	0.02	0.02

Table 5.2: Comparison of NSCF and DOS input parameters

5.10 Precautions

- Ensure SCF calculation has converged before running NSCF
- Use same `outdir` and `prefix` in all files
- For DOS, use `occupations = 'tetrahedra'` for accurate results
- Set appropriate `emin` and `emax` to capture all relevant states
- Fermi level should be at 0 eV in DOS plot

Chapter 6

Lab 7: Electronic Band Structure Calculation

6.1 Aim

To calculate and visualize the electronic band structure of Copper using Quantum ESPRESSO .

6.2 Theory

6.2.1 Band Structure

The electronic band structure describes the relationship between electron energy and crystal momentum (E vs $\mathbf{k}$) in a solid. It reveals:

- Whether a material is a metal, semiconductor, or insulator
- Effective mass of electrons and holes
- Electronic properties (conductivity, optical transitions)

Key Concepts

- **Brillouin Zone**: The primitive cell in reciprocal space
- **High-symmetry points**: G (Γ), X, W, L, K, etc.
- **Band gap**: Energy difference between valence and conduction bands
- **Fermi energy**: Energy level at $T = 0$ K
- For Cu (FCC): High-symmetry path is **G-X-W-L-G-K**

6.2.2 Software Used

- **XCrySDen**: For visualizing Brillouin zone and selecting k-path
- **xmgrace**: For plotting band structure

- **Quantum ESPRESSO**: For SCF and bands calculations

6.3 Connection to Previous Experiments

Complete Workflow

Chapter 4: ECUT & k-point Convergence
↓
Chapter 5: SCF Calculation
↓
Chapter 6: NSCF & DOS Calculations
↓
Chapter 7: Band Structure Calculation

6.4 Step 1: SCF Calculation for Primitive Cu

For band structure, we use the **primitive cell** of Cu (1 atom per cell) instead of the conventional cell.

SCF Input File for Primitive Cu (`scf.in`)

```
&CONTROL
    calculation = 'scf'
    etot_conv_thr = 1.0000000000d-05
    forc_conv_thr = 1.0000000000d-04
    outdir = './out/'
    prefix = 'aiida'
    pseudo_dir = './'
    tprnfor = .true.
    tstress = .true.
    verbosity = 'high'
/
&SYSTEM
    degauss = 2.0000000000d-02
    ecutrho = 7.2000000000d+02
    ecutwfc = 9.0000000000d+01
    ibrav = 0
    nat = 1
    nosym = .false.
    ntyp = 1
    occupations = 'smearing'
    smearing = 'cold'
/
&ELECTRONS
    conv_thr = 2.0000000000d-10
    electron_maxstep = 80
    mixing_beta = 4.0000000000d-01
/
ATOMIC_SPECIES
    Cu 63.546 Cu.pbe-dn-kjpaw_psl.1.0.0.UPF
ATOMIC_POSITIONS crystal
    Cu 0.0000000000 0.0000000000 0.0000000000
K_POINTS automatic
    21 21 21 0 0 0
CELL_PARAMETERS angstrom
    2.5296254900 0.0000000000 0.0000000000
    1.2648127450 2.1907199364 0.0000000000
    1.2648127450 0.7302399788 2.0654305636
```

Key Differences from Conventional Cell

- `nat = 1` (primitive cell has only 1 Cu atom)
- `CELL_PARAMETERS` are primitive vectors (not cubic)
- `K_POINTS = 21 21 21` (dense grid for accurate SCF)

6.5 Step 2: Visualizing Brillouin Zone with XCrySDen

How to Find k-path using XCrySDen

1. Open the SCF input file in XCrySDen:

```
xcrysden --pwi scf.in
```

2. Go to **Tools** → **Brillouin Zone**
3. Select **k-path** for FCC structure
4. The standard path for FCC is: **G-X-W-L-G-K**
5. Note down the coordinates of each k-point

Point	Symbol	Coordinates
Gamma	G	(0.000, 0.000, 0.000)
X	X	(0.000, 0.500, 0.500)
W	W	(-0.250, 0.250, 0.500)
L	L	(-0.500, 0.000, 0.000)
Gamma	G	(0.000, 0.000, 0.000)
K	K	(-0.375, 0.000, 0.375)

Table 6.1: High-symmetry k-points for FCC structure (Cu)

6.6 Step 3: Band Structure Calculation

Band Structure Input File (bands.in)

```
&CONTROL
    calculation = 'bands'          ! Band structure calculation
    etot_conv_thr = 1.0000000000d-05
    forc_conv_thr = 1.0000000000d-04
    outdir = './out/'
    prefix = 'aiida'
    pseudo_dir = './'
    tprnfor = .true.
    tstress = .true.
    verbosity = 'high'
/
&SYSTEM
    degauss = 2.0000000000d-02
    ecutrho = 7.2000000000d+02
    ecutwfc = 9.0000000000d+01
    ibrav = 0
    nat = 1
    nosym = .false.
    ntyp = 1
    occupations = 'smearing'
    smearing = 'cold'
    nbnd = 15                      ! Include empty bands
/
&ELECTRONS
    conv_thr = 2.0000000000d-10
    electron_maxstep = 80
    mixing_beta = 4.0000000000d-01
/
ATOMIC_SPECIES
    Cu 63.546 Cu.pbe-dn-kjpaw_psl.1.0.0.UPF
ATOMIC_POSITIONS crystal
    Cu 0.0000000000 0.0000000000 0.0000000000
K_POINTS {crystal_b}               ! Path in Brillouin zone
    6
    0.0000000000 0.0000000000 0.0000000000 33 ! G
    0.0000000000 0.5000000000 0.5000000000 33 ! X
    -0.2500000000 0.2500000000 0.5000000000 33 ! W
    -0.5000000000 0.0000000000 0.0000000000 33 ! L
    0.0000000000 0.0000000000 0.0000000000 33 ! G
    -0.3750000000 0.0000000000 0.3750000000 33 ! K
CELL_PARAMETERS angstrom
    2.5296254900 0.0000000000 0.0000000000
    1.2648127450 2.1907199364 0.0000000000
    1.2648127450 0.7302399788 2.0654305636
```

Key Parameters

- `calculation = 'bands'` (instead of 'scf' or 'nscf')
- `K_POINTS {crystal_b}` defines the path
- The number after each point (33) is the number of points along the segment
- `nbnd = 15` includes empty bands for visualization

6.7 Step 4: Post-Processing for Band Structure

Post-Processing Input (pp.in)

```
&BANDS
    prefix = 'aiida'
    outdir = './out/'
    filband = 'copper.dat'
/
```

Running the Commands

```
# Step 1: Run SCF
pw.x < scf.in > scf.out

# Step 2: Run Bands calculation
pw.x < bands.in > bands.out

# Step 3: Run Post-processing
bands.x < pp.in > pp.out
```

6.8 Step 5: Plotting Band Structure

Plotting with xmgrace

1. Open the data file:

   ```
   xmgrace copper.dat
   ```

2. The vertical axis shows energy (eV)
3. The horizontal axis shows k-points along the path
4. Mark the high-symmetry points (G, X, W, L, G, K)

Electronic Band Structure of Copper

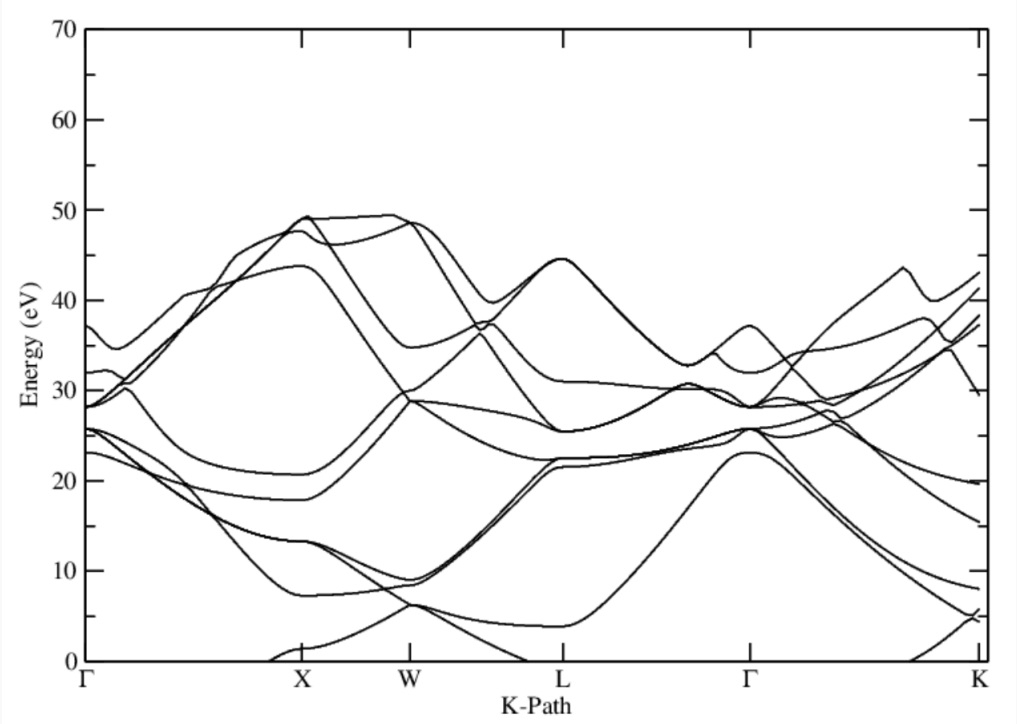


Figure 6.1: Band structure of Copper (FCC) along G-X-W-L-G-K path

Reading the Band Structure

- **Fermi energy**: Red dashed line at E_F
- For Cu (metal): Bands cross the Fermi level
- The bands below E_F are occupied (valence bands)
- The bands above E_F are empty (conduction bands)
- The overlapping bands indicate metallic behavior

6.9 Summary of Files

File	Command	Purpose
scf.in	pw.x ¡ scf.in ¿ scf.out	Ground state density
bands.in	pw.x ¡ bands.in ¿ bands.out	Band structure calculation
pp.in	bands.x ¡ pp.in ¿ pp.out	Extract band data
copper.dat	xmgrace copper.dat	Plot band structure

Table 6.2: Summary of files for band structure calculation

6.10 Important Notes

FCC High-Symmetry Points

- **G (Γ)**: (0, 0, 0) - Center of Brillouin zone
- **X**: (0, 0.5, 0.5) - Face center point
- **W**: (-0.25, 0.25, 0.5) - Corner point
- **L**: (-0.5, 0, 0) - Body center point
- **K**: (-0.375, 0, 0.375) - Special point

6.11 Precautions

- Use primitive cell for band structure (not conventional)
- Ensure `outdir` and `prefix` are consistent
- SCF calculation must be run before bands calculation
- Include sufficient empty bands (`nbnd`)
- Verify k-point coordinates from XCrySDen
- Use `K_POINTS {crystal_b}` for band path

Appendix a: VESTA Reference Guide

The following reference manual provides essential instructions for using VESTA software.

Introduction

When crystal structures are analyzed by you, discussed with fellow researchers or provided in scientific publications, some sort of visual representation as needed. A solely text-based communication of crystallographic data is in most cases incomprehensible.



©2000 Howard Hughes Medical Institute. Image may not be distributed or further reproduced without permission.

In the early days of crystallography these representations of crystal structures were drawn by hand. This very extensive method is still being used to create stunning pictures like this painting of a cytochrome. However, it took even very skilled artists like Irving Geis up to several months to complete a picture of a crystal structure.

Computer Programs for the visual representation of crystal structures may not match the artistic aspects of Irving Geis' watercolor paintings. But they allow a very rapid display of crystallographic information files on your computer, tablet or even your smartphone. In this tutorial we want to show you some very basic techniques on the handling of crystal structures in such a computer program:

The creation of a unit cell from given lattice parameters
The addition of several atoms into this unit cell
The packing of several unit cells into a super cell

We will be using a 3D visualization program, called VESTA. It can display structural models as well as crystal morphologies. It is distributed free of charge for academic, scientific, educational, and noncommercial users. It is available for most operating systems (Mac OS X, Linux and Windows.)

This tutorial covers the installation process on Windows 8.1.

You can either use this written tutorial or watch the corresponding video. In both cases you need the PDF as a source of the crystallographic information.

If you wish to skip the process of transcribing a crystal structure and just want to try VESTA on your own, there is a ready-made CIF as additional resource.

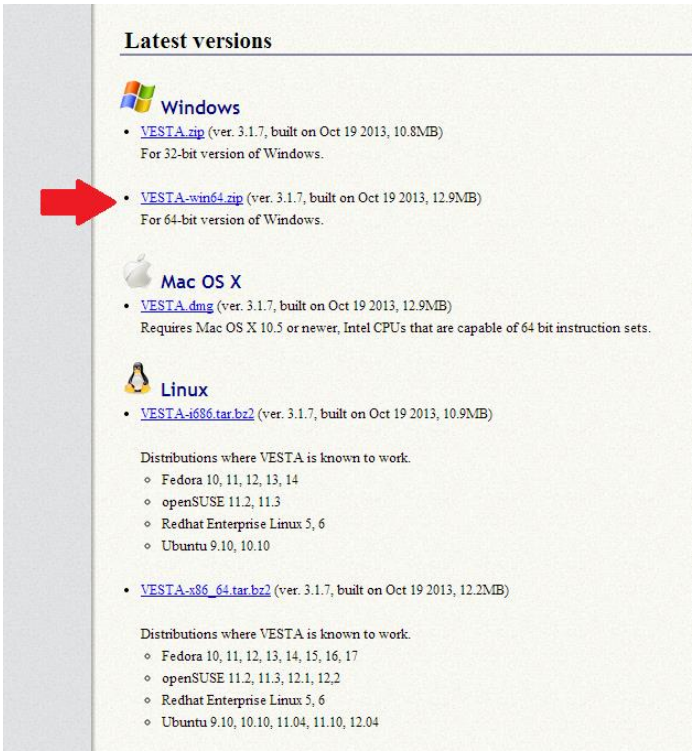
Installation

Go to: http://jp-minerals.org/vesta/en/ and click on **Downloads**.

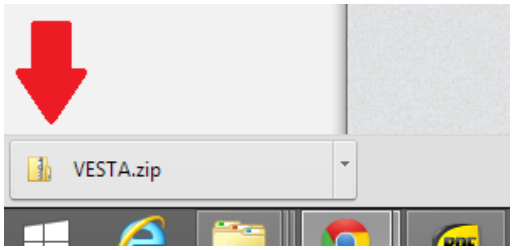


Download the program files. I downloaded the 64 bit Version since I run the appropriate Operating System.

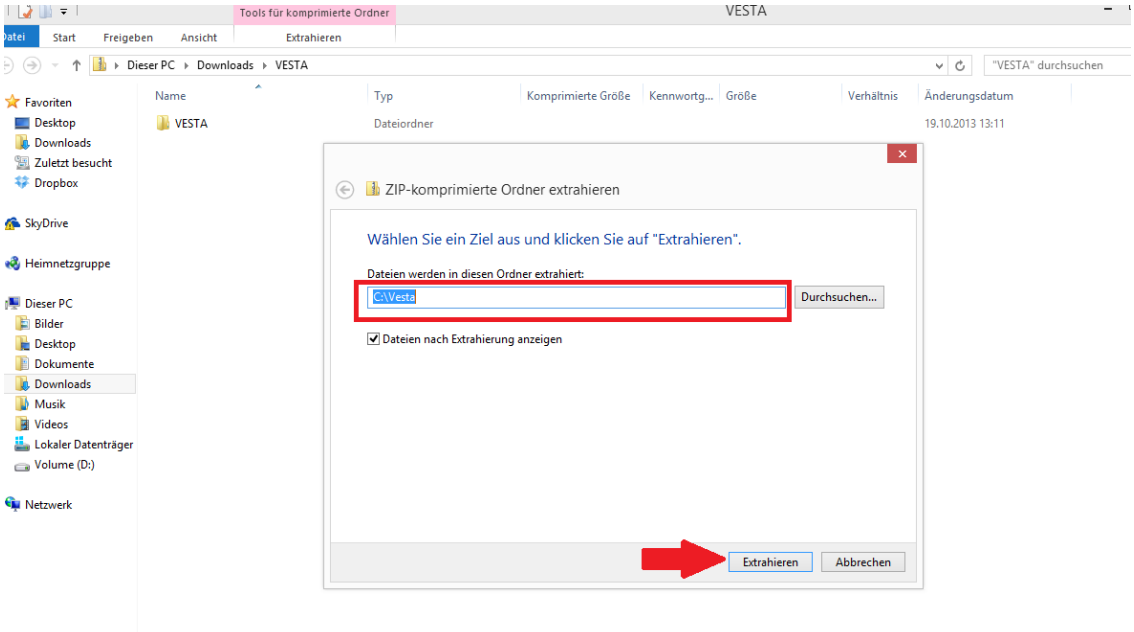
(If you are unsure whether you need the 32 or 64 bit version, check your type of installation of Windows. You can do this by hitting the windows button, then type system, enter. This will launch this window here. If you find this is too complicated, just take the 32 bit version.)



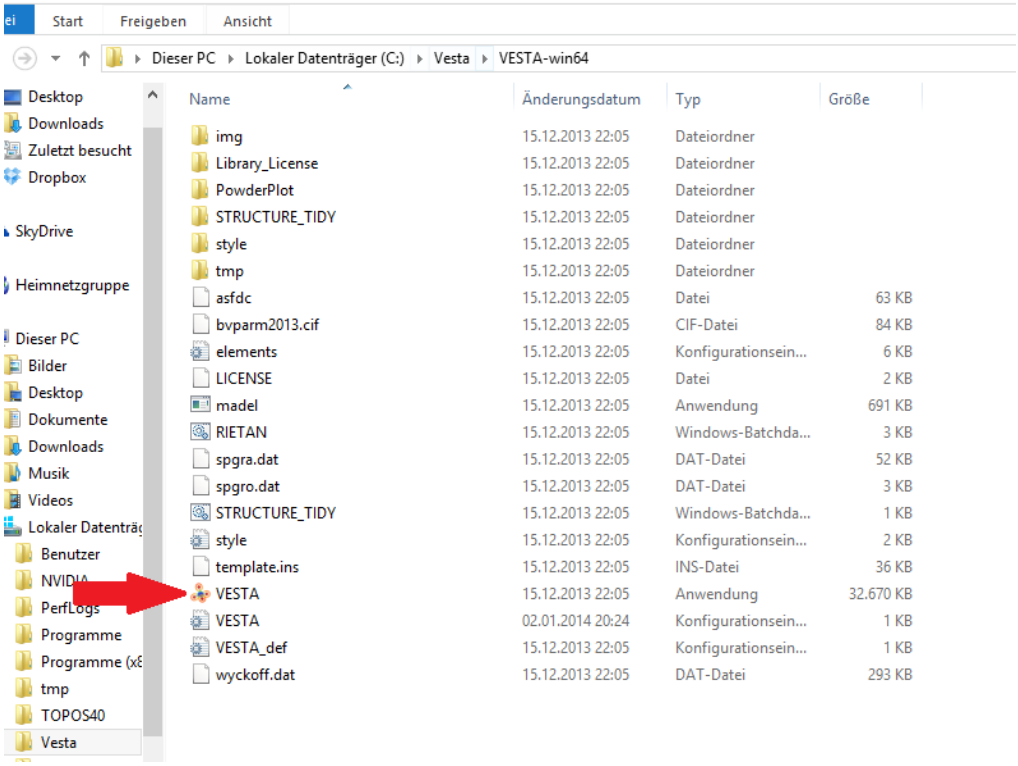
In your download folder you have a ZIP file



Extract the ZIP file to a location on your hard drive. I chose C:\Vesta



This extracted folder contains everything you need. There is no further installation required. Search this folder for an application called VESTA.

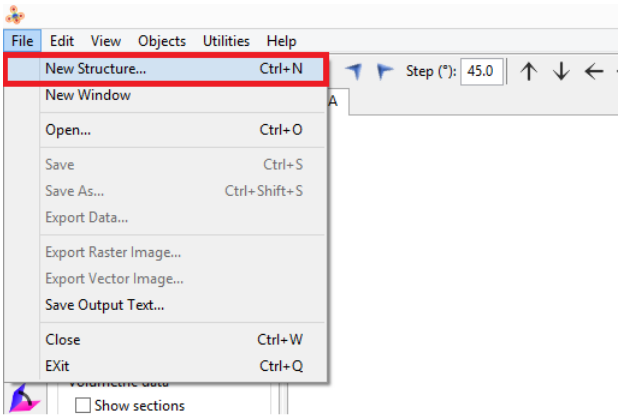


Double click it and you are running VESTA.

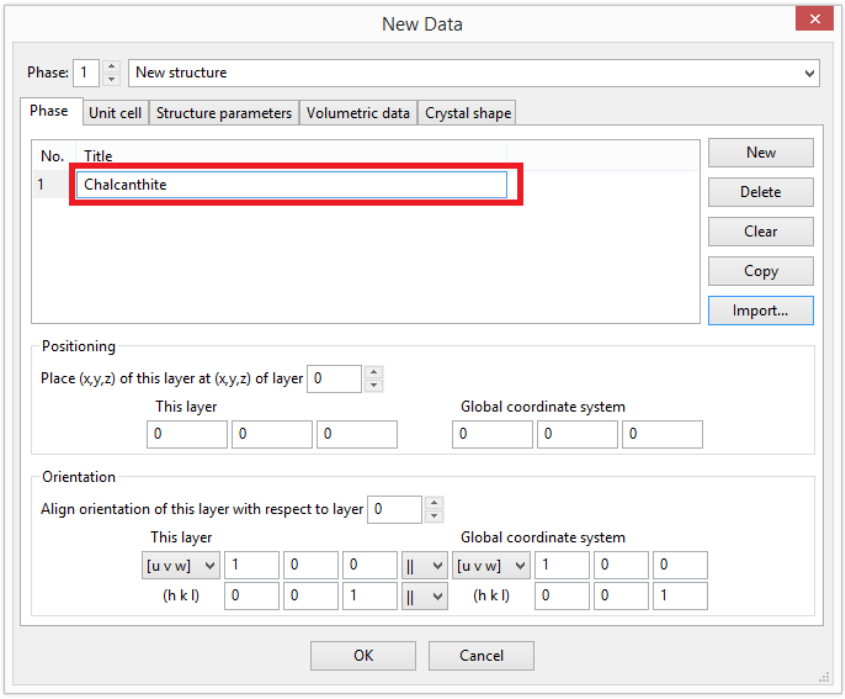
Create a new Structure

We want to build a crystal structure just from the unit cell dimensions and fractional co-ordinates of the atoms within this cell. You find this data in the PDF in the additional resources.

Click File – New Structure



By default the new structure has a generic name. **Click** the Name Field to change this to something more descriptive i.e. Chalcanthite.

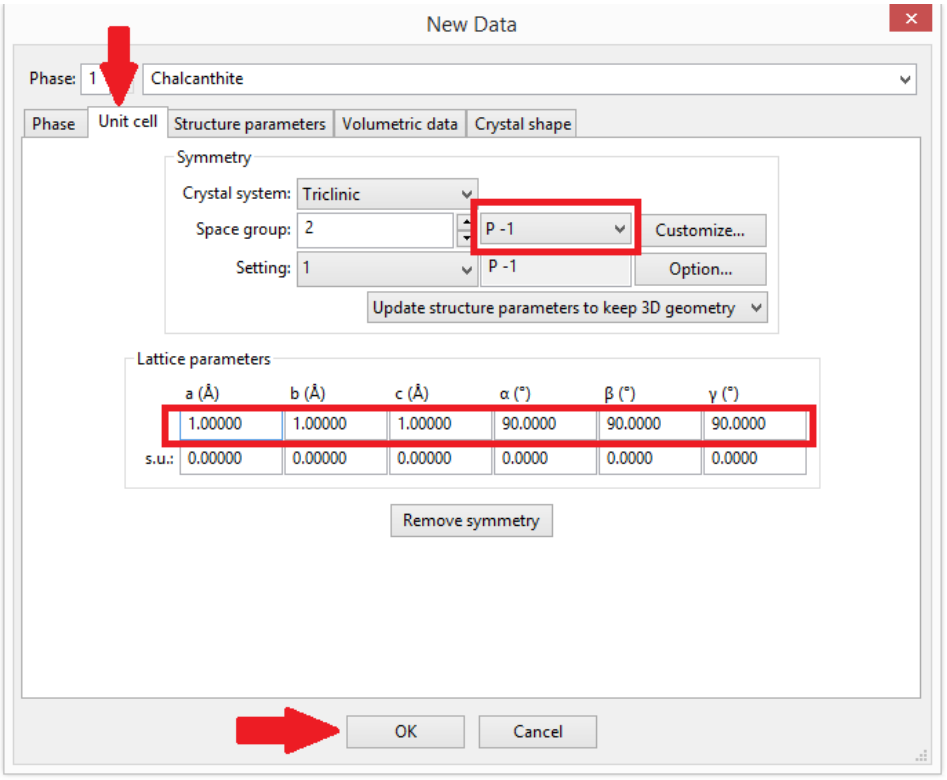


Switch to the **Unit Cell Tab**.

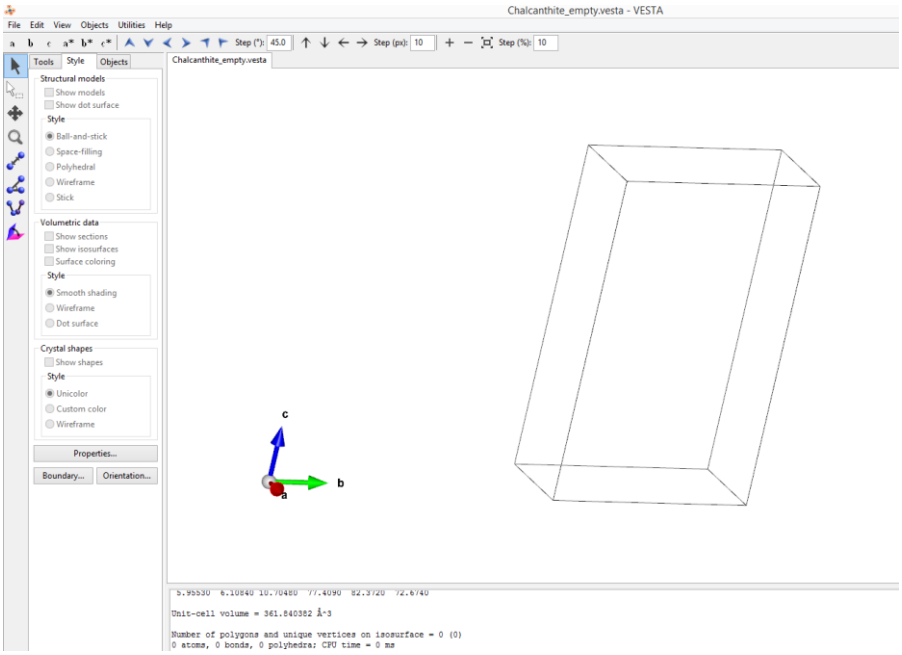
First, you need to change the space group. Space groups describe the symmetry elements that can be applied to the unit cell. We will elaborate on this later in this course. For now, just **change the space group to P-1** (it is a space group with an inversion center).

Enter the lattice parameters as given in the PDF.

Click OK to save your input.

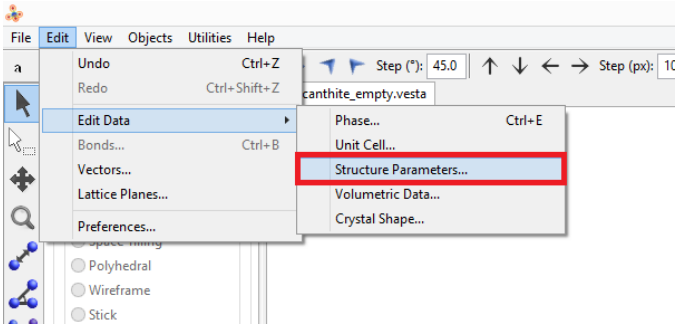


A unit cell has been created. It should look something like this. Click and drag your cursor over the unit cell to view it from different angles.

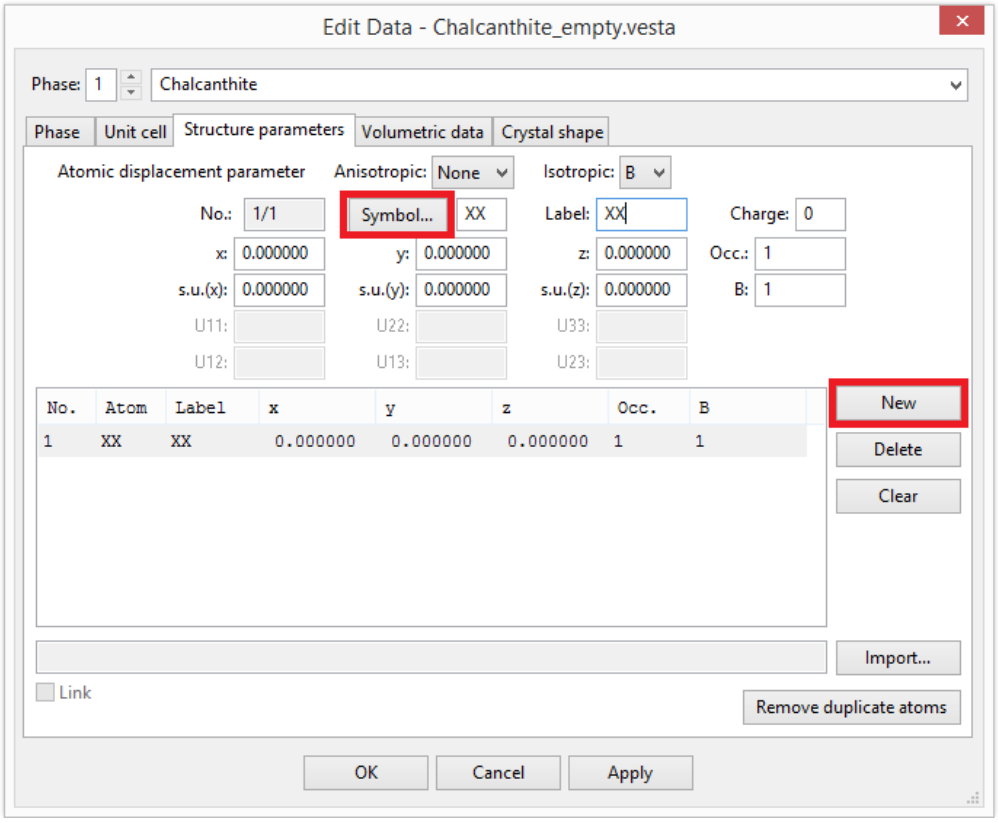


Let's add some atoms to the unit cell.

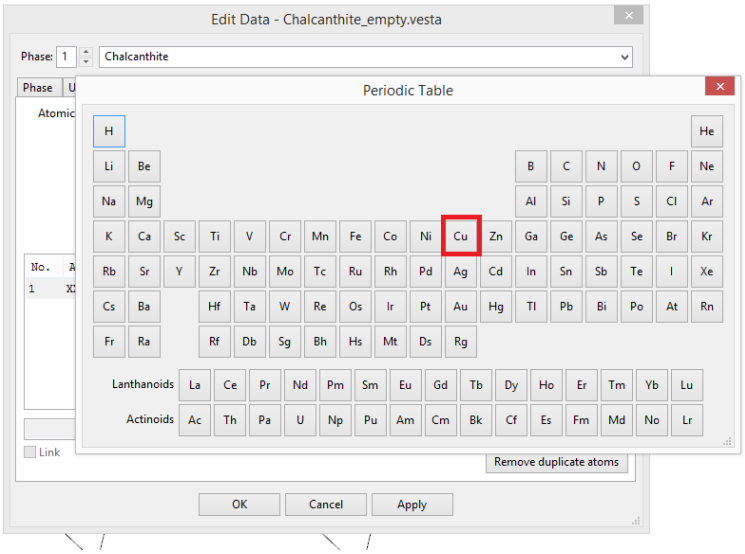
In the top Menu click **Edit – Edit Data – Structure Parameters**



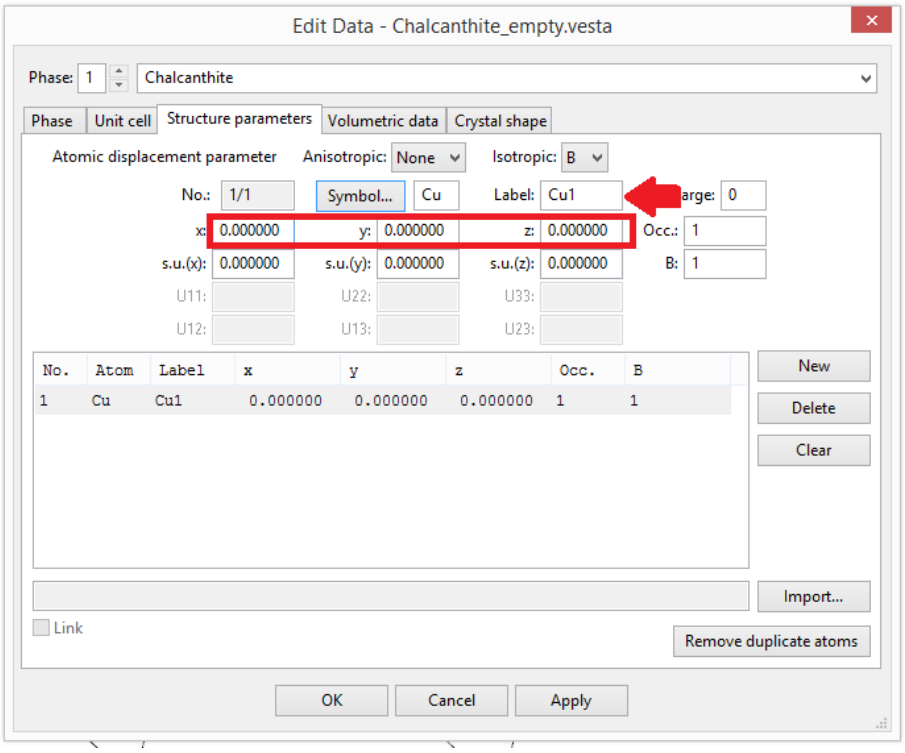
In the structure parameters tab you can add or delete atoms from your crystal structure. Click **New** to add a new atom then click **Symbol** to choose an element from the periodic table.



A window with the periodic table pops up. Choose **Copper** as your first atom.



This atom needs a label to be distinguished from other copper atoms that are present in this structure in a later stage. Although you are free to label the atom with anything you want, we suggest **Cu1** to label it unambiguously.

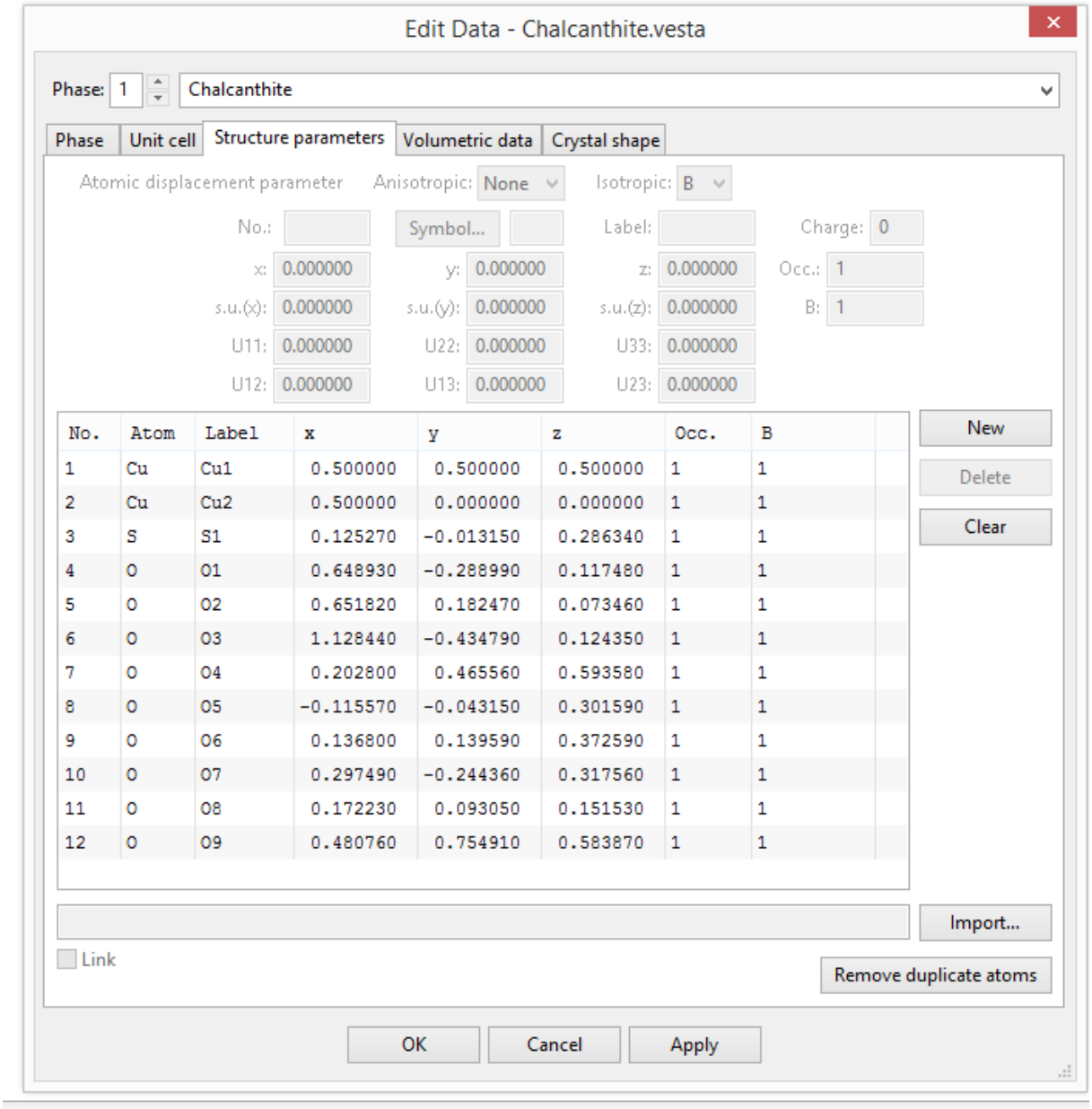


Enter the fractional coordinates of the first copper atom from the pdf into the respective fields of x, y and z. Please note that in the pdf the coordinates are given as ½ , ½ and ½ . However, they you should enter them as 0.5 into VESTA.

Click **Apply** or **OK** save to enter your input.

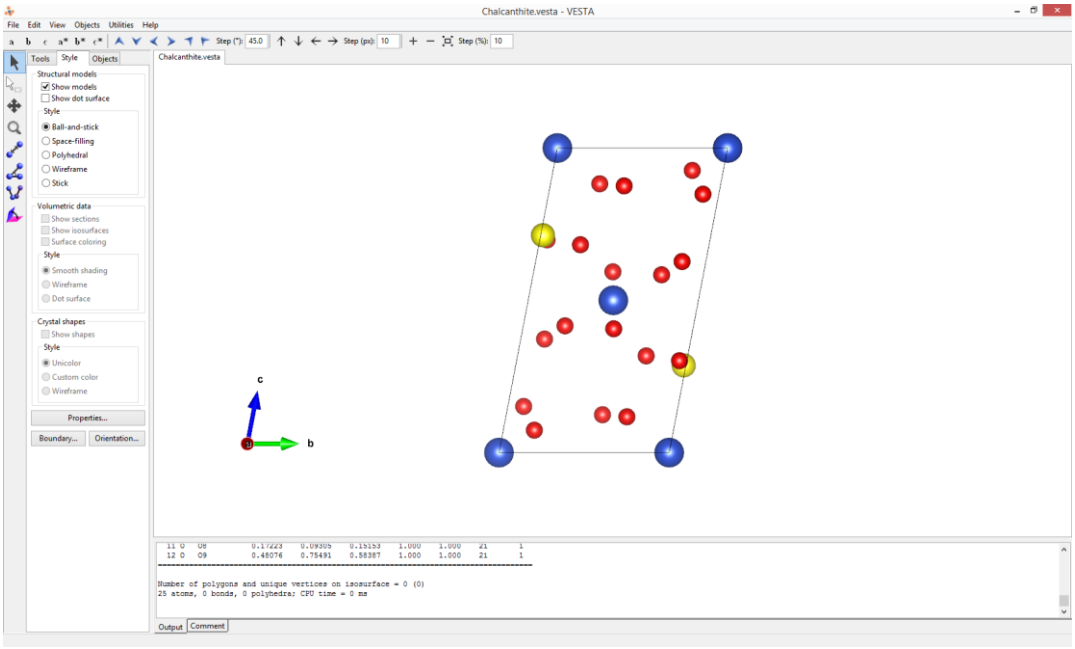
There are a total of twelve atoms to be added to the crystal structure. Repeat the steps described above for the other atoms. Do not forget to change the element symbol when entering the coordinates for sulfur and oxygen. Also make sure that you enter the negative values for some of the coordinates correctly.

Every atom entered this way is listed in the structure parameters tab. When complete it should look like this.



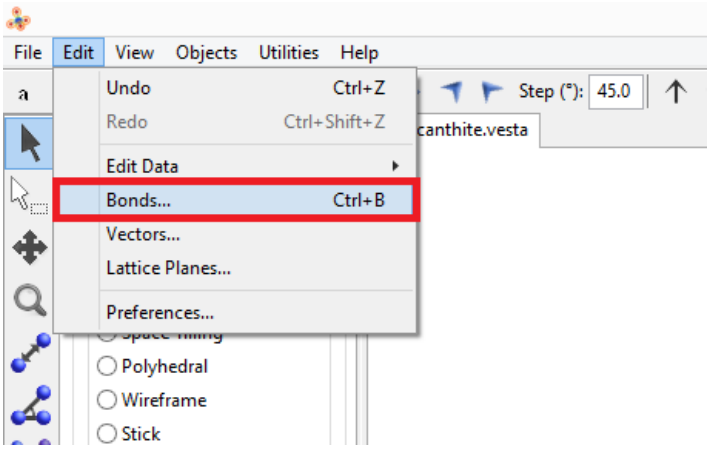
Click **OK** to save.

VESTA is now displaying a full unit cell.



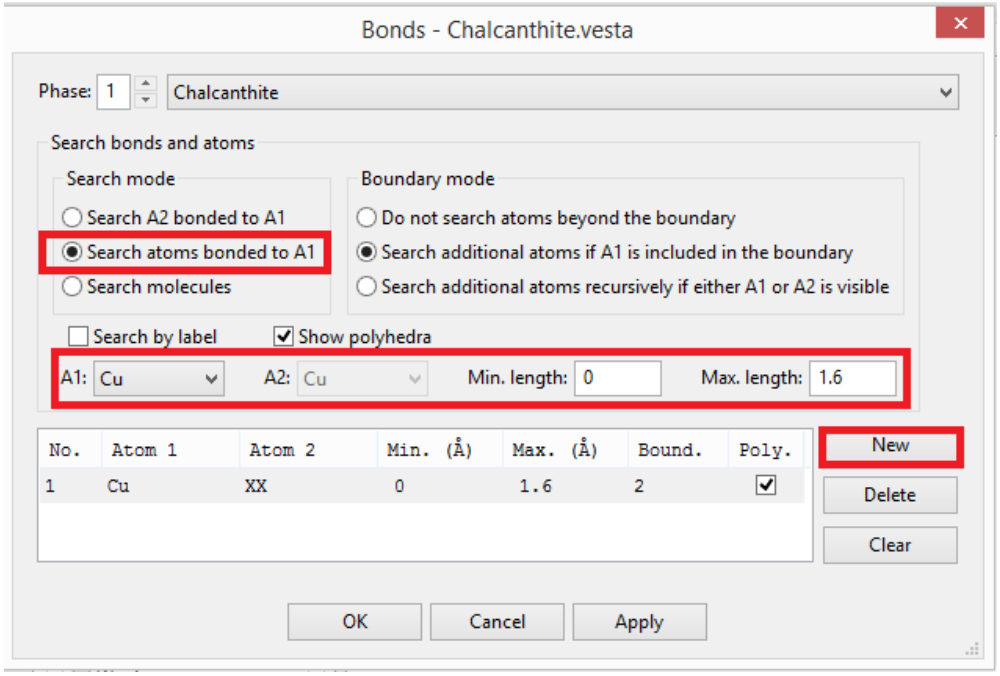
But there are no bonds between the atoms. Let's change that.

In the Top Menu click **Edit - Bonds**

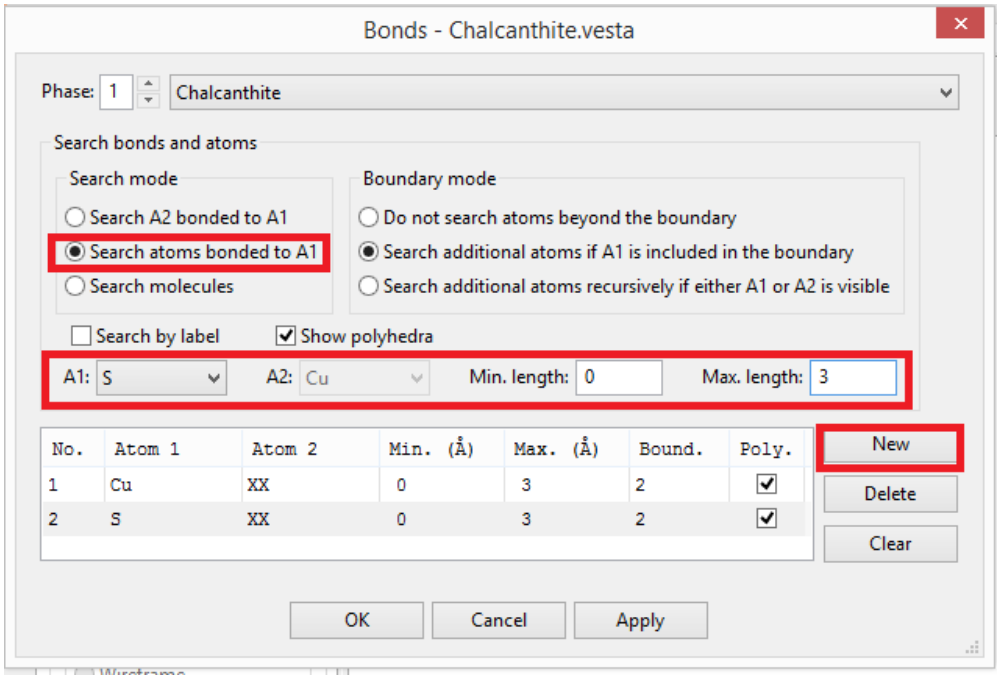


We want to create a new set of bonds for all copper atoms. Click **New**. Choose the option **Search atoms bonded to A1**. Set A1 to Cu.

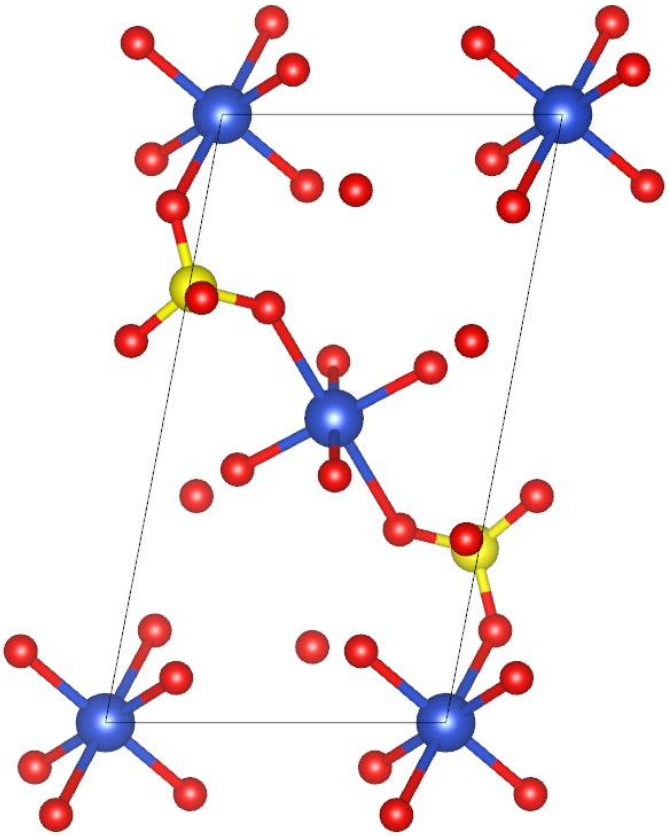
With this option being set, VESTA searches all bonds to a given atom within a given range. By default VESTA searches for bonds to a set type of atoms within 0-1.6 Å. This bonding range won't result in any bonds. So it needs to be increased a little. **Set the maximum length** to 3 Å. Click OK or Apply to save your input.



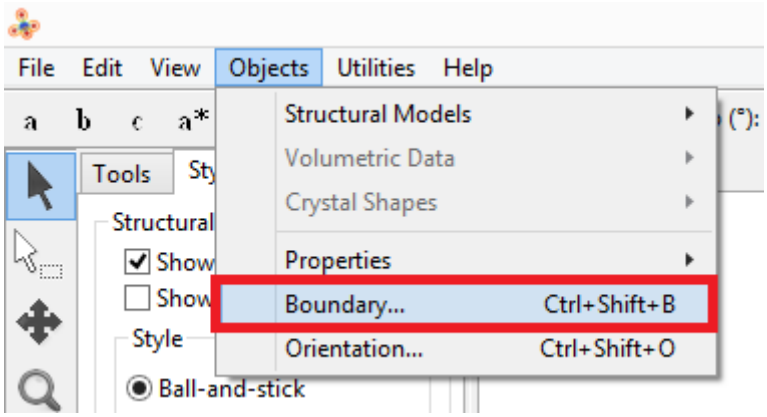
Repeat this process for Sulfur.



Now all the copper and sulfur atoms are bonded to oxygen. Look at the structure from different angles and observe the difference in the bonding environment between copper and sulfur.

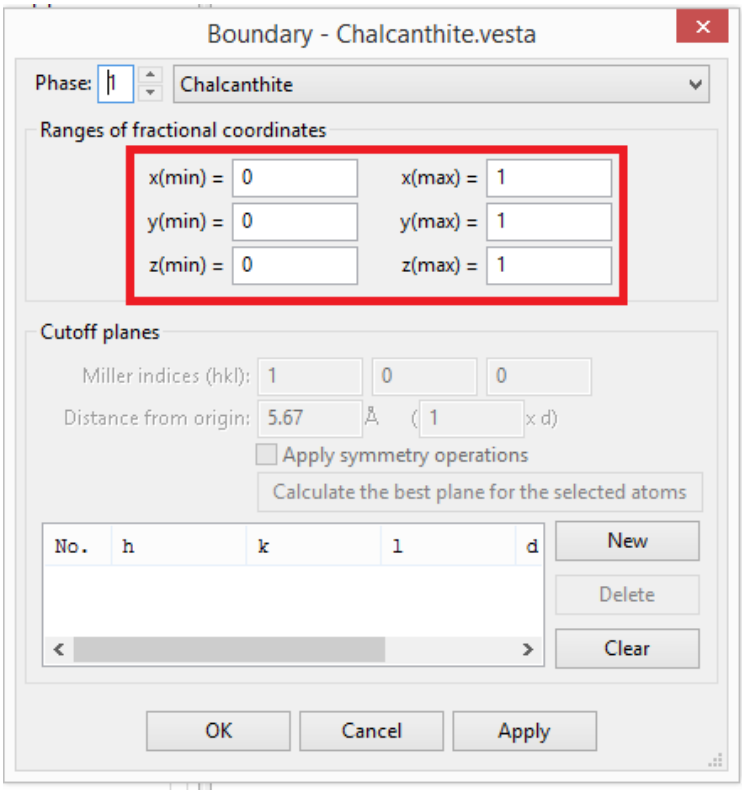


Now let's display more than just one unit cell.
In the Top Menu click **Objects – Boundary.**

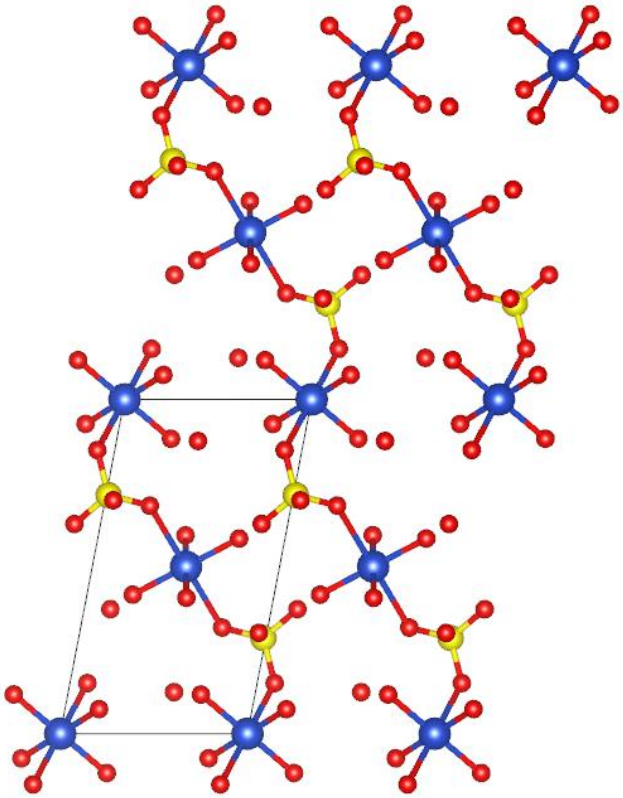


The default settings set to display all atoms that are located between the fractional coordinates from 0-1 for the x, y and z-direction. Atoms outside of these coordinates are displayed too if they are bonded to atoms that lie within these coordinates (This can be disabled in the boundary mode setting in the **Edit – Bonds** tab).

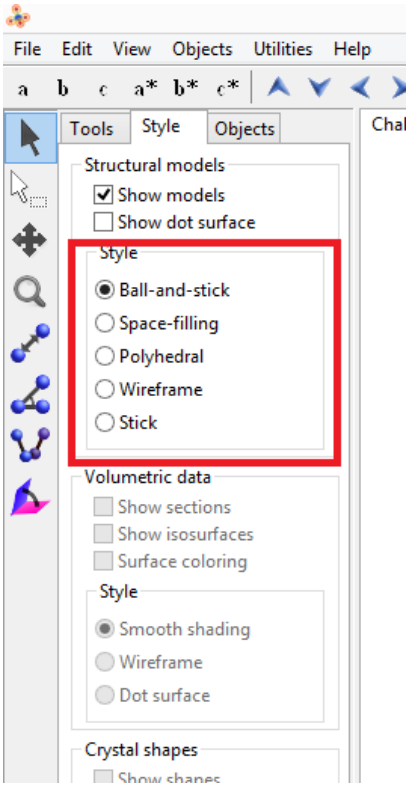
To display a so-called super cell, set the **x(max)** setting to 2. Repeat this setting for **y(max)** and **z(max)**. Click **OK** to save your input.



Now a super cell is displayed. It consists of eight individual unit cells. This arrangement can actually be called a crystal as it shows a periodic arrangement of atoms. However this crystal would be very, very tiny.



Congratulations! You have created your first virtual crystal in VESTA. We suggest that you use this structure now to play around with VESTA's different **display types**. Try for example a **space-filling** representation or the **polyhedral** mode.

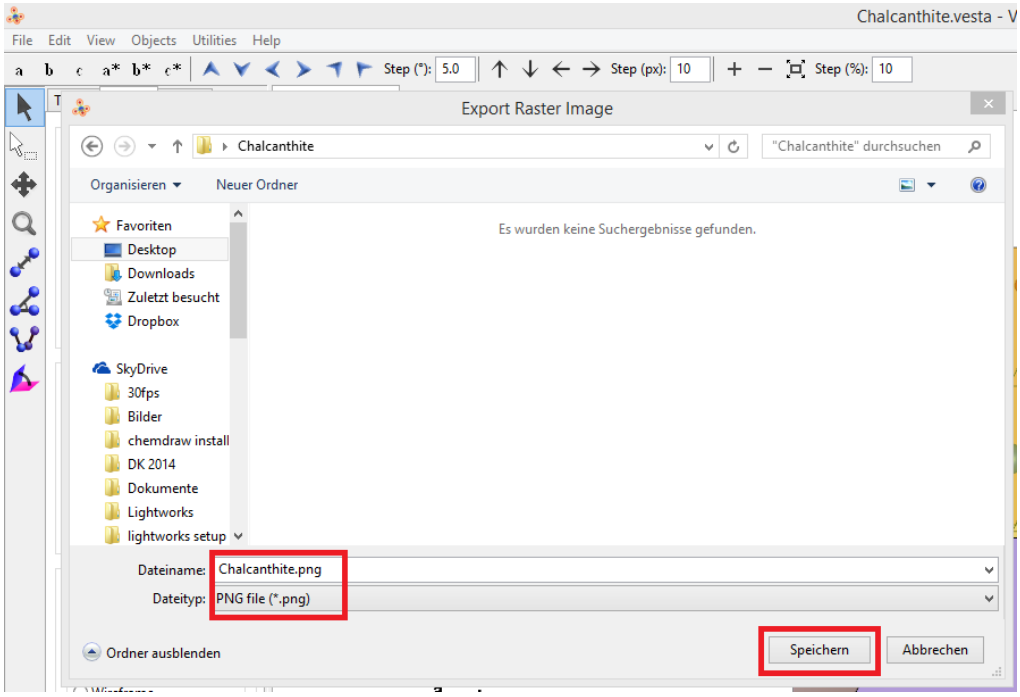


Don't forget to **save** the file you have created.

You can do this by clicking: **File, Save As** and the choosing the **VESTA file format**. And what you can also do is to **export** your file as a **crystallographic information file**. Click **File, Export** and then choose the **CIF** option to export as a CIF file.

Share Your Pictures

You can save them as a PNG or a JPG file by clicking **File >> Export Raster image.**



Appendix b : Sample Input Files for Quantum ESPRESSO

6.12 1. Relaxation Input File (`Fe_relax.in`)

Relaxation Calculation - Fe BCC

```
&CONTROL
    calculation = 'relax'          ! Type of calculation: relax, scf, nscf, vc-relax
    prefix = 'Fe'                  ! Prefix for output files
    outdir = './outputs/'          ! Directory for temporary files
    pseudo_dir = './pseudo/'       ! Directory containing pseudopotentials
    forc_conv_thr = 1e-4           ! Force convergence threshold (Ry/au)
/

&SYSTEM
    ibrav = 0                      ! 0 = read cell from CELL_PARAMETERS
    nat = 2                        ! Number of atoms
    ntyp = 1                       ! Number of atom types
    ecutwfc = 50                   ! Plane-wave cutoff energy (Ry)
    ecutrho = 400                  ! Charge density cutoff (4-8 times ecutwfc)
/

&ELECTRONS
    conv_thr = 1e-8                ! SCF convergence threshold
    mixing_beta = 0.7              ! Mixing factor for electron density
/

&IONS
    ion_dynamics = 'bfgs'          ! Optimization algorithm: bfgs, damp, sd
/

ATOMIC_SPECIES
    Fe  55.845   Fe.pbe-spn-kjpaw.UPF   ! Mass and pseudopotential file

CELL_PARAMETERS {angstrom}
    2.87  0.00  0.00
    0.00  2.87  0.00
    0.00  0.00  2.87

ATOMIC_POSITIONS {crystal}
    Fe  0.00  0.00  0.00
    Fe  0.50  0.50  0.50

K_POINTS {automatic}
    8  8  8   0  0  0              ! Monkhorst-Pack k-point grid
```

Important Tags for Relaxation

- `calculation = 'relax'` - Performs ionic relaxation (atoms move to minimize forces)
- `calculation = 'vc-relax'` - Relaxes both atoms AND cell parameters
- `forc_conv_thr` - Forces below this value indicate convergence
- `ion_dynamics = 'bfgs'` - Broyden-Fletcher-Goldfarb-Shanno optimization algorithm

6.13 2. SCF Input File (`Fe_scf.in`)

SCF Calculation - Fe BCC

```
&CONTROL
    calculation = 'scf'            ! Single point SCF calculation
    prefix = 'Fe'
    outdir = './outputs/'
    pseudo_dir = './pseudo/'
    verbosity = 'high'             ! Detailed output for debugging
/

&SYSTEM
    ibrav = 0
    nat = 2
    ntyp = 1
    ecutwfc = 50
    ecutrho = 400
    occupations = 'smearing'       ! For metals: smearing, for insulators: fixed
    smearing = 'gaussian'          ! Smearing type: gaussian, mp, fermi
    degauss = 0.02                 ! Smearing width (Ry)
    nspin = 1                      ! 1 = no spin, 2 = spin-polarized
/

&ELECTRONS
    conv_thr = 1e-8                ! SCF convergence threshold (Ry)
    mixing_beta = 0.7
    electron_maxstep = 100         ! Maximum SCF iterations
    diagonalization = 'david'      ! Davidson diagonalization algorithm
/

ATOMIC_SPECIES
    Fe  55.845   Fe.pbe-spn-kjpaw.UPF

CELL_PARAMETERS {angstrom}
    2.87  0.00  0.00
    0.00  2.87  0.00
    0.00  0.00  2.87

ATOMIC_POSITIONS {crystal}
    Fe  0.00  0.00  0.00
    Fe  0.50  0.50  0.50

K_POINTS {automatic}
    12  12  12   0  0  0          ! Denser grid for accurate energy
```

Important Tags for SCF

- `calculation = 'scf'` - Performs only electronic SCF (no ionic motion)
- `conv_thr` - Energy convergence threshold (smaller = more accurate)
- `occupations` - `'smearing'` for metals, `'fixed'` for insulators
- `degauss` - Smearing width (helps metallic systems converge)

6.14 3. NSCF Input File (`Fe_nscf.in`)

NSCF Calculation for Band Structure

```
&CONTROL
    calculation = 'nscf'          ! Non-SCF calculation
    prefix = 'Fe'
    outdir = './outputs/'
    pseudo_dir = './pseudo/'
/

&SYSTEM
    ibrav = 0
    nat = 2
    ntyp = 1
    ecutwfc = 50
    ecutrho = 400
    occupations = 'smearing'
    smearing = 'gaussian'
    degauss = 0.02
    nbnd = 20                     ! Number of bands to calculate (more than needed)
/

&ELECTRONS
    conv_thr = 1e-8
    diagonalization = 'david'
/

ATOMIC_SPECIES
    Fe  55.845   Fe.pbe-spn-kjpaw.UPF

CELL_PARAMETERS {angstrom}
    2.87  0.00  0.00
    0.00  2.87  0.00
    0.00  0.00  2.87

ATOMIC_POSITIONS {crystal}
    Fe  0.00  0.00  0.00
    Fe  0.50  0.50  0.50

K_POINTS {crystal_b}              ! Path for band structure
    20
    0.000  0.000  0.000   20      ! Gamma point
    0.500  0.000  0.500   20      ! X point
    0.500  0.250  0.750   20      ! W point
    0.375  0.375  0.750   20      ! K point
    0.000  0.000  0.000   1       ! Gamma point
```

Important Tags for NSCF & K-Point Path

- `calculation = 'nscf'` - Non-SCF: uses potential from previous SCF run
- `nbnd` - Number of bands (include empty bands for accurate DOS)
- `K_POINTS {crystal_b}` - Defines path in Brillouin zone for band structure
- Format: `kx ky kz weight` - Points along high-symmetry directions

6.14.1 Comparison Table of Input Types

Parameter	Relaxation	SCF	NSCF
`calculation`	relax / vc-relax	scf	nscf
`k-points`	6x6x6 to 8x8x8	8x8x8 to 12x12x12	Path for bands
`conv_thr`	1e-6 to 1e-8	1e-8 to 1e-10	1e-8
`forc_conv_thr`	1e-3 to 1e-4	Not needed	Not needed
`ion_dynamics`	bfgs	Not needed	Not needed
`nbnd`	Default	Default	Set higher
Purpose	Find stable structure	Ground state energy	Band structure/DOS

Table 6.3: Comparison of relaxation, SCF, and NSCF input parameters

6.14.2 Common K-Point Paths for Different Structures

Crystal System	High-Symmetry Path	Points
BCC (Fe)	G-H-N-G-P	G=(0,0,0), H=(0.5,0.5,0.5), N=(0,0.5,0)
FCC (Cu)	G-X-W-K-G-L-U-W	G=(0,0,0), X=(0.5,0,0.5)
Hexagonal (Graphene)	G-M-K-G	G=(0,0,0), M=(0.5,0,0), K=(0.33,0.33,0)

Table 6.4: Common k-point paths for band structure calculations

6.14.3 How to Run These Calculations

Run Commands

```
# Step 1: Run Relaxation (if needed)
pw.x < Fe_relax.in > Fe_relax.out

# Step 2: Run SCF (always required)
pw.x < Fe_scf.in > Fe_scf.out

# Step 3: Run NSCF (for bands/DOS)
pw.x < Fe_nscf.in > Fe_nscf.out
```

Important Note

For NSCF calculation, ensure you have already run an SCF calculation in the same `outdir`. The NSCF code reads the charge density from the SCF run!